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Effect of DDT on the protein and free amino acids concentration of the ovary, fat bodies and muscles in adult female, *Dysdercus cingulatus* (Hemiptera : Pyrrhocoridae)

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Abstract : Effect of DDT on the protein and amino acid concentration of the ovary, fat bodies and muscles in adult female *Dysdercus cingulatus* was investigated in the laboratory under normal condition as well as experimental. The concentration of protein of ovary, fat bodies and muscles of the insect declined significantly in case of DDT treated insect in comparison to control insect. Decline in protein concentration corresponds to increase in the free amino acids at respective stages of treated insect. In the adult female *Dysdercus cingulatus*, Dichloro Diphenyl Trichloroethane (DDT) had lowering effect on protein concentration of the ovary, Fat bodies and muscles from one day to 7 days. The effect of DDT on the free amino acid concentration of the ovary, fat bodies and muscles of the adult female *Dysdercus cingulatus* from one day to three day had decreased and significantly increased from 4th day to 7th day.

Key words: Protein, Free amino acid, DDT, Ovary Fat bodies, Muscles and *Dysdercus cingulatus*.

INTRODUCTION

Dysdercus cingulatus is a fast breeding cotton pest such as the number increase to a considerable magnitude. If proper control over their population is not applied, they increase to a great number and cause a lot of damage to the cotton plant and vegetable plants. The red cotton nymphs were conspicuous and quite accessible to the action of insecticides than other stages of the adult insects.

Effect of DDT on the protein and amino acid concentration of the ovary, fat bodies and muscles in mature adult female *Dysdercus cingulatus* were estimated in present study. Estimation of protein and free amino acids concentration has been qualitatively and quantitatively carried out by various workers on terrestrial insects and aquatic insects under various conditions. Guthrie (1950)[1] has described the application of DDT at low temperature in *Blatella germanica*, Wyatt (1961)[2] in insect haemolymph, Islam and Roy (1982; 1983)[3,4] in *Schizodactylus monstrosus* Roy *et al.* (1983)[5] in chrysocoris, Phukon *et. al.* (1986)[6] in *Hydrophilous olivaceous* Kumari and Naidu (1987)[7] in *Bombyx mori* Rajani *et al.* (1989)[8] in the tissues of fresh water Prawn, *Caridina rajadhari*, Satya Sharma *et al.* (1988)[9] in *Sphaerodema rusticum* and Rani *et al.* (1991)[10] in *Hydrophilous olivaceous* (Fab.) Boggs *et al.* (1991)[11] described the effect of DDT on the physiology of Butterfly, Raju and Gopalan (1998)[12] described the effect of insecticides on the body of rice leaf folder eggs. Singh

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(2002)[13] has studied on the effect of insecticides on adult of red cotton bug. Vyas (2009)[14] and Patnayak (2012)[15] have observed the effect of endosulfan on the body of *D. koenigii* Hassan *et al.* (2014)[16] investigated the relative toxicity of insecticides on the nymphs of *Chrysocoris stolli*. Mala and David (2016)[17] have described the ovicidal effect of certain insecticides to the pumpkin caterpillar. However, little information is available with regard to concentration of protein and free amino acids under different condition with DDT treated insects.

MATERIALS & METHODS

The adult *Dysdercus cingulatus* were collected from cotton plants and reared in the laboratory under controlled condition (30 ± 2^0C and 70 ± 5% RH). The known ages (one day old) *Dysdercus cingulatus* were selected and divided into two groups. One group (60 to 70) of insects reared in cage of laboratory at normal condition i.e. provided food (seeds of cotton) without the application of any chemicals and served as control. The other group (60 to 70) of insects was exposed to the sub lethal dose of DDT (0.0025%). The newly emerged adults were acclimatized atleast for two times prior to performing experiments. The concentration of Protein and free amino acids were estimated after one day, 2nd day, 3rd day, 4th day, 5th day, 6th day and 7th day of DDT treated insects. The quantitative estimation of ovary, fat bodies and muscles protein as well as free amino acids of adult female *Dysdercus cingulatus* were done by Lowry *et al.* (1951)[18] and Moore & Stein (1954)[19] methods respectively.

RESULTS & DISCUSSION

Estimation of protein and free amino acids concentration of different tissues of the mature female *Dysdercus cingulatus* were studied under following parameters:

1. PROTEIN

OVARY: The ovary of untreated mature female *Dysdercus cingulatus* had increased level of protein concentration from 1 day to 7 days (Table-1). Protein concentration had 84.20 mg/g in one day 90.00 mg/g in 2nd day, 93.20 mg/g in 3rd day, 95.00 mg/g in 4th day, 96.50 mg/g 5th day, 97.20 mg/g in 6th day and 98.00 mg/g in 7th day old untreated female *Dysdercus cingulatus* (Graph-1).

The ovary of DDT treated female *Dysdercus cingulatus* had decreased level of protein concentration from 1 day to 7 days (Table-1). Protein concentration had 84.20 mg/g, in 1 day, 78.50 mg/g in 2nd day, 66.50 mg/g in 3rd day, 54.20 mg/g in 4th day, 38.40 mg/g 5th day, 22.50 mg/g in 6th day and 12.00 mg/g in 7th day old treated female *Dysdercus cingulatus* (Graph-2)

FAT BODIES

Fat bodies of untreated adult female insect had also increased level of protein concentration from one day to 7th day. Protein concentration had 50.40 mg/g in 1 day, 55.20 mg/g in 2nd day, 60.20 mg/g in 3rd day, 66.00 mg/g in 4th day, 70.00 mg/g in 5th day, 72.50 mg/g in 6th day and 75.20 mg/g in 7th day old insects (Table-1).

In DDT treated female, the fat body protein concentration declined to 50.40, 42.00, 36.00, 30.50, 26.40, 18.50 and 10.00 mg/g ($P<0.001$) from one day old to 7th day old (Graph-1).

MUSCLES

The female *Dysdercus cingulatus* had increased level of muscles protein concentration from one day old to seven days old (Table-1). The protein concentration of muscles had 90.00 mg/g in 1 day, 91.50 mg/g in 2nd, 93.00 mg/g in 3rd day, 94.20 mg/g in 4th day, 96.50 mg/g in 5th day, 97.30 mg/g in 6th and 98.00 mg/g in 7th day old female *Dysdercus cingulatus*.

In DDT treated female, the protein concentration of muscle had declined from one day to 7th days. The protein concentration muscles had 90.00 mg/g in 1 day, 87.00 in 2nd day, 66.00 in 3rd day, 48.00 in 4th day, 35.50 mg/g in 5th day, 20.00 mg/g in 6th day and 10.00 mg/g ($P < 0.001$) in 7th day old female *Dysdercus cingulatus* (Graph-2).

2. FREE AMINO ACIDS

OVARY

The concentration of free amino acids in ovary of untreated female adult *Dysdercus cingulatus* had slowly increased from one day to 7th day (Table-2). Free amino acid concentration of ovary had 65.00 mg/g in 1 day, 67.20 mg/g in 2nd day, 67.50 mg/g in 3rd day & 4th day, 68.00 mg/g found in 5th day & 6th day and 68.50 mg/g presented in 7th day old female insect (Graph-3).

In the DDT treated female, the concentration of free amino acids of ovary had slowly decreased from 1 day to 3rd day but gradually increased from 4th day to 7th day. Free amino acids concentration are found to be 63.00 mg/

g in one day, 65.00 mg/g in 2nd day, 66.50 mg/g in 3rd day, 70.00 mg/g in 4th day, 75.00 mg/g in 5th day, 82.00 mg/g in 6th day and 90.00 mg/g in 7th day old female *Dysdercus cingulatus* (Graph-4).

FATBODIES

The fat body of untreated adult female had slowly increased free amino acids level from 1-day to 7th day. Free amino acids level had 62.50 mg/g in 1 day, 63.50 in 2nd day 64.00 in 3rd day, 64.50 mg/g in 4th day and 5th day but 65.00 mg/g in 7th day old female insect (Table-2).

In DDT treated female, the free amino acid concentration of fat bodies had slowly increased in 1-day to 3rd day but significantly increased from 4th day to 7th day (Graph-3). The free amino acids had 60.20 mg /g in one day, 62.50 mg/g in 2nd day, 63.50 mg/g in 3rd day, 72.00 mg/g in 4th day, 78.00 mg/g in 5th day, 85.00 mg/g in 6th day and 92.00 mg/g in 7th day old female, *Dysdercus cingulatus*.

MUSCLES

The untreated adult female *Dysdercus cingulatus* had the muscles free amino acids concentration of 70.20 mg/g in one day, 72.00 mg/g in 2nd day, 73.50 mg/g in 3rd day, 74.00 mg/g in 4th day, 74.20 mg/g in 5th day, 74.50 mg/g in 6th day and 75.00 mg/g in 7th day old (Table-2). In DDT treated female insect, free amino acids concentration of muscles had slowly decrease in one day to 3rd day and significantly increased in age of 4th day to 7th day. (Graph-4). The free amino acids had investigated 68.50 mg/g in age of one day 67.00 mg/g in 2nd day and 66.20 mg/g in 3rd day but 78.50 mg/g in age of 4th day, 82.00 mg/g in 5th day, 88.50 mg/g ($P < 0.001$) in 6th day and 96.00 mg/g in age of 7th day old treated female *Dysdercus cingulatus*.

Decline in the body protein concentration and increases in the free amino acids level had described by Islam and Roy (1982), Roy *et al.* (1983) in *Schizodactylus monstrosus*. Kumari and Naidu (1987) described the effect of Permethrin in *Bombyx mori.*

Present investigation on adult female *Dysdercus cingulatus*, DDT had significantly decreased the protein concentration in ovary, fat bodies and muscles from one day to seven days. So, DDT had affected the protein synthesizing machinery of the adult female insects. DDT had decreased activity in free amino acids from one day to 3rd day old muscles of insect but significantly increased from the age of 4th day to 7th day old muscles of female insects. This is in according with the finding of Rajni *et al*. (1989), and Rani *et al.* (1991) in *Hydrophilous olivaceous* (Fab).

Table 1-Protein concentration in different tissues of adult female, *Dysdercus cingulatus* (mg/g)

S. N.	Tissues	1st day		2nd day		3rd day		4th day		5th day		6th day		7th day	
		UT	T	UT	T	UT	T	UT	T	UT	T	UT	T	UT	T
1	Ovary	84.20	84.20	90	78.50	93	66.50	95	54.20	96.50	38.40	97.20	22.50	98	12
2	Fat bodies	50.40	50.40	55.20	42	66.20	36	66	30.50	70	26.40	72.50	18.50	75.20	10
3	Muscles	90	90	91.50	87	93	66	94.20	48	96.50	35.50	97.30	20	98	10

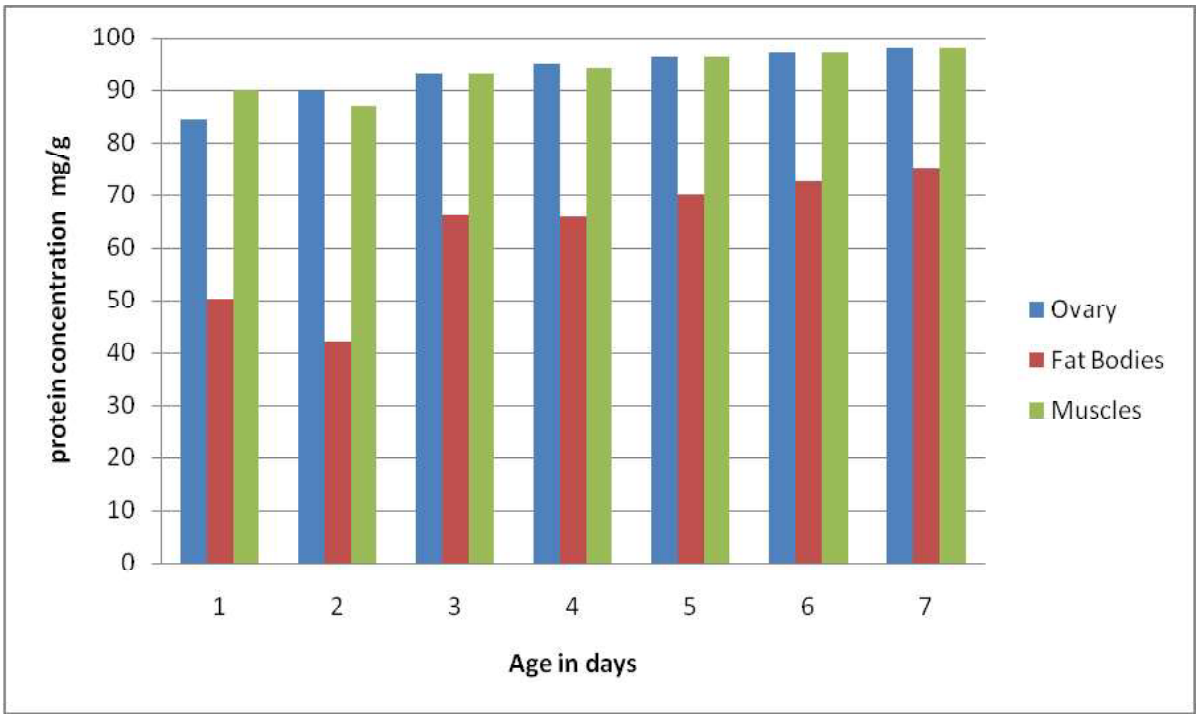
UT- Untreated insect, T- DDT treated insect

Table 2- Free amino acids concentration in different tissues of adult female, *Dysdercus cingulatus* (mg/g)

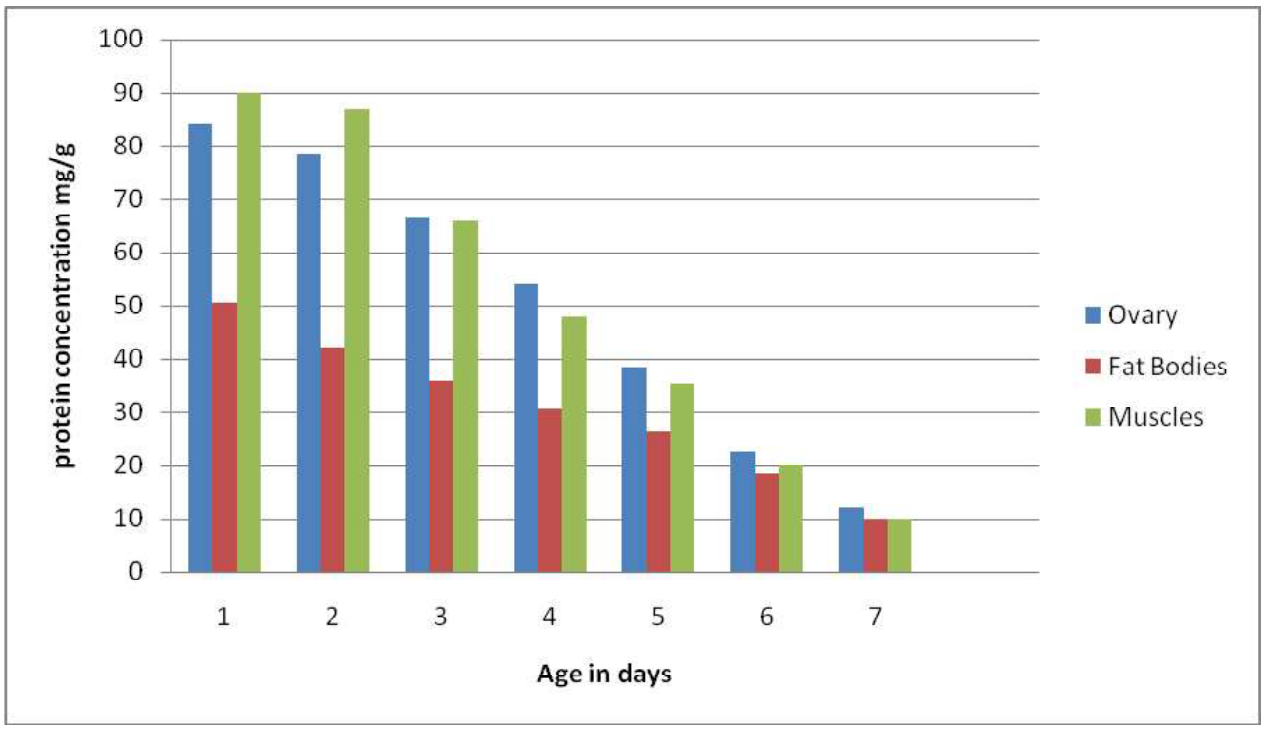
S. N.	Tissues	1st day		2nd day		3rd day		4th day		5th day		6th day		7th day	
		UT	T	UT	T	UT	T	UT	T	UT	T	UT	T	UT	T
1	Ovary	65	63	67.20	65	67.50	66.50	67.50	70	68	75	68	82	68.50	90
2	Fat bodies	62.50	60.20	63.50	62.50	64	63.50	64.50	72	64.50	78	64.50	85	65	92
3	Muscles	70.20	68.50	72	67	73.50	66.20	74	78.50	74.20	82	74.50	88.50	75	96

UT- Untreated insect, T- DDT treated insect

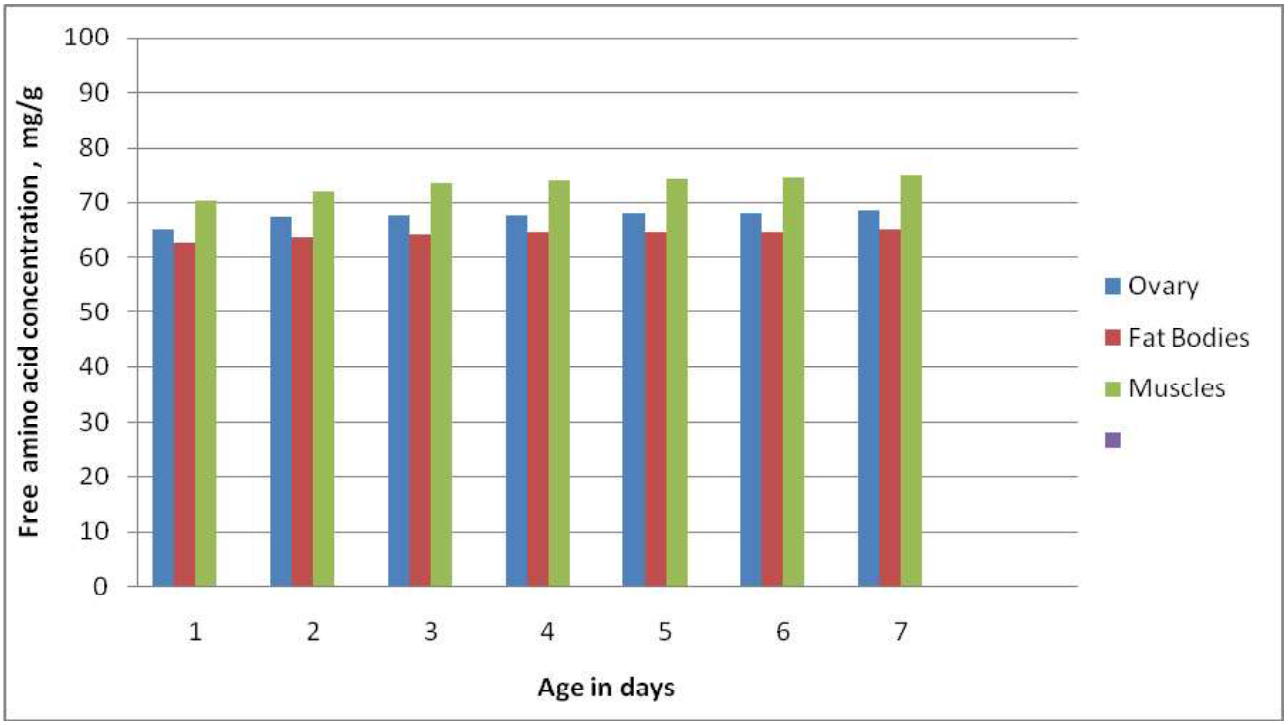
Graph 1 - Graph showing protein concentration in different tissues of untreated (UT) adult female, *Dysdercus cingulatus*.



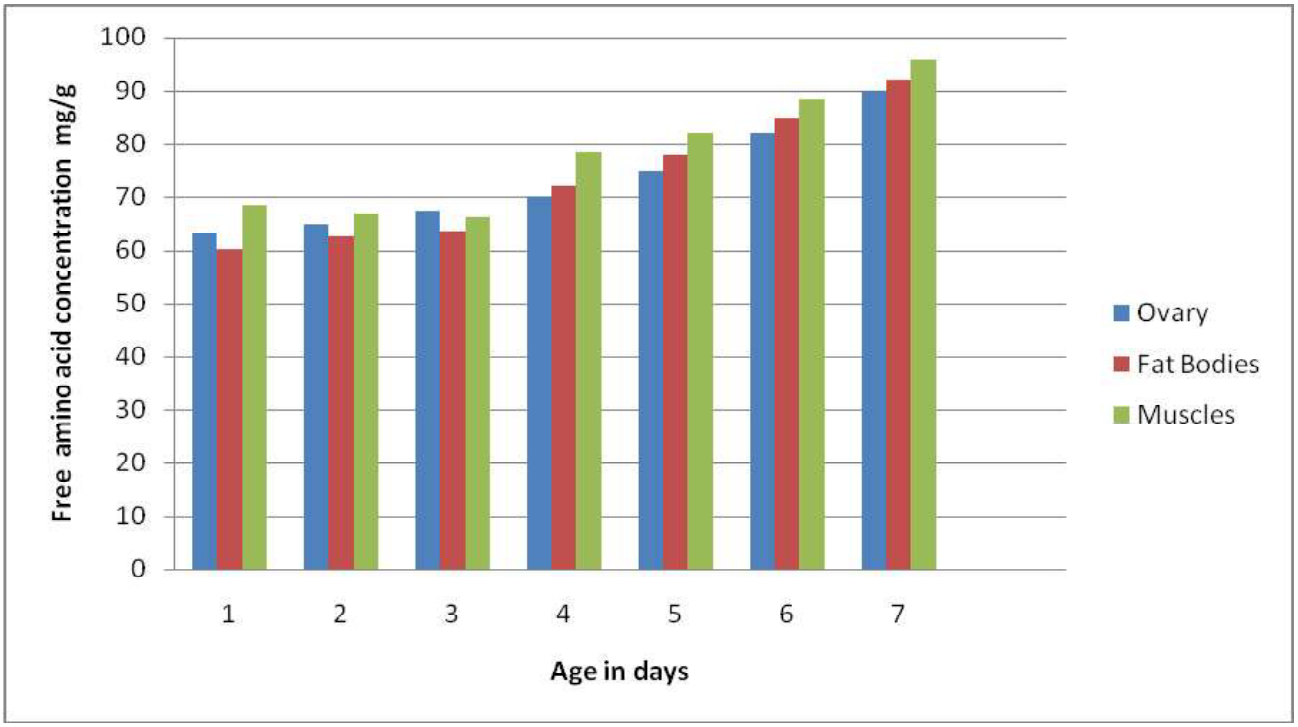
Graph 2 - Graph showing protein concentration in different tissues of treated (T) adult female, *Dysdercus cingulatus*.



Graph 3- Graph showing free amino acids concentration in different tissues of untreated (UT) adult female, *Dysdercus cingulatus.*



Graph 4- Graph showing free amino acids concentration in different tissues of treated (T) adult female, *Dysdercus cingulatus*.



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Endohelminthic parasite diversity in piscine host of Madhepura district, Bihar, India

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Abstract : Fishes constitute a major component of diet for the people of India and they are extensively used as a protein rich food for human consumption. The present study incorporates the diversity of the parasitic species in freshwater fishes of different water bodies from Madhepura with a view to reduce the endohelminthic infestation and protect the piscine host as healthy food resource. The diversity spectrum in the form of relative abundance and S-W diversity indices of endohelminth parasites recovered from the various piscine host species in the present study has been dealt in th elight of protecting the host form the parasites.

Key words: Endohelminths, piscine host, parasite diversity indices, protection strategy.

INTRODUCTION

Helminths are an important group of pathogens, which cause infection and diseases of fish both in freshwater and marine environments, their importance being related directly to the fish that may affect the general public health (Hoffman, 1967)[1]. As much as 30,000 helminth species have been estimated to be parasites of fishes, many of which are known to be serious menace to their hosts. Every parasite living in or on a fish extends some degree of harmful influence on its host (Williams and Jones 1994)[2]. Understanding patterns in species diversity and distribution, and evaluating the role of biodiversity in ecosystem function, remain key goals for contemporary ecologists (Sutherland *et al.* 2006)[3]. Parasitism is one of the most successful modes of life displayed by living organisms, as measured by how often it evolved and how many parasitic species are presently in existence. Studying the diversity of parasites is particularly relevant because sympatric diversification may be important in some parasite taxa, and because of the opportunity for independent tests of evolutionary hypotheses in the many separate lineages in which parasitism evolved. Patterns in the diversity of parasites may be associated with either host or parasite characteristics. The distribution of parasite diversity among host taxa does not simply reflect the species diversity of the host taxa themselves; life history and ecological traits of hosts appear to play important roles. These may determine the likelihood that hosts are colonized by parasite species over evolutionary time. Certain features of parasites may also be associated with speciation and diversification. Only parasite body size has received much attention; the patterns observed are not greatly different from those of free-living species, with small-bodied parasite taxa being more species than related large-bodied taxa. The emphasis for future research must shift from pattern description to the elucidation of the processes responsible

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for the structure and diversity of parasite faunas. A better integration of ecological and historical (or phylogenetic) approaches to the study of parasite diversity should make this objective possible. Species diversity or richness of fish tapeworms appears to be greatest in subtropical and temperate regions. Nematodes of fish infective to humans are mostly in the tropics.

MATERIALS & METHODS

The fishes were collected and analyzed to study the parasitic diversity. The sampled fishes were analyzed in the laboratory for the presence of endohelminthic parasites. The parasites were identified and indexed. Study sites are rivers and ponds of Madhepura district in Bihar. The freshly landed fishes *Carassius auratus* and *Cirrhinus cirrosus* were collected from fish landings of Madhepura district of Bihar coast (lat. 25.8674° N and Long 86.9424°´E) in winter season. Samples were transferred to departmental laboratory aquarium. An internet search, using the keywords 'host parasite checklist' and the search engine Google, was conducted to find relevant checklists; additional searches were conducted by replacing 'host' with 'fish', and 'parasite' with 'helminth'. Only endohelminth parasites (nematodes) are considered here. Finally, only two fish host species with at least seven endoparasite species in the surveyed region were included. To observe the helminth parasites fishes were opened up vertically and the entire digestive system was removed and placed in a Petri dish with physiological saline. Fishes were identified by using the key[4,5]. Parasites were identified[6,7,8]. The external and internal organs were thoroughly examined for parasitic load. The parasites were mainly found in the gastrointestinal tract. The GIT specifically stomach and intestine was removed carefully and kept separately in Petri dish. All the parasites recovered from the different organs of gastrointestinal tract were preserved in 70% alcohol. They were cleaned and kept in small vials for the study of their morphology for identification with the help of light microscope Olympus BX43. The parasites number and place of their attachment were also recorded.

Nematodes- Nematodes were identified by the naked eye or by using lens. Specimens were washed in water or in a physiological solution and fixed in a 5% formalin Nematodes of smaller dimensions after fixation were cleared using a drop of phenol to elucidate the structures such as esophagus and oral spicules for species identifications besides the body shape and dimensions the characters such as the body cuticle, existence of spines the structure of the anterior and posterior body are important. It is also necessary to know the host and location of the nematode. After the microscopic examination the nematodes were carefully rinsed in water and transferred again into a fixative [9].

RESULT & DISCUSSION

A total of 143 fishes of *Carassius auratus* and *Cirrhinus cirrosus* were screened occurring abundantly in the Madhepura district The present study recorded nematodes belonging to the families Ascaridae, (anisakidae). In the present investigation besides the fish hosts various parasite was identified as *Chilodonella hexastica, Trichodina subtilis, Trichodina reticulate, Apiosoma piscicola, myxobolus mrigalhite, myxobolus basui,* and *myxobolus lalbaghensis*. Population of the two fishes was chosen for the helminth parasitic faunal study in Madhepura waters. Among the endohelminths recorded *Myxobolus basui* was observed to be the highest infectious parasite in *Cirrhinus cirrhosus* species. And the lowest was Apiosoma piscicola infecting the *Carassius auratus.* The site of infection was in the Gill lamellae especially in the intestine harboured maximum number of endohelminthic parasites. From the two host genera examined *Cirrhinus cirrosus* was found to be highly infected with nematode parasites in comparison to *Carassius auratus* . From all over parasitic burden 45 prasitic species have been reported from liver of *Cirrhinus cirrosus* and intestine have been reported in *Carassius auratus* as shown in table 1. It is clear from Table 1, that prevalence and infection was observed . Body of edible fishes is an environment to the parasites. Due to parasitic diseases, the growth of fishes decreases and almost ceases. Das and Goswami (2014)[10] reported that each helminths parasite species prefer to live in a definite zone of the microhabitats, though some can migrate to the other organs, which are normally not their usual site of infection. Devi *et al*., (2015)[11] have studied that many parasite species are host specific to at least some degree and are capable of infecting one or only a limited number of host species. Rukhsana *et al.*, (2008)[12] reported that due to several factors these fishes are declining

in number at a very fast rate from last few years. Chowdhury and Hossain (2015)[13] also observed that parasitic infestation has harmful influence for fish health that affects the normal growth of the fishes and mortalities as effect of the helminth parasites in terms of loss of body weight and mortality was also reported in our study. Fish frequently serve as intermediate or transport hosts for larval parasites. Wide spectrum of helminths infections is transmitted to humans by contamination of food and water as reported by Panda and Dash (2016)[14]. Ajit and Yuvraj (2015)[15] reported that fish health maintenance programs require many elements, including examination and monitoring of fish pathogens. Some of these parasites cause diseases to fish, affecting their health and reproduction, making them fall easy prey to predators and some infect man. Wali *et al.*, (2016)[16] also reported that fish carrying heavy parasitic burden are extremely lethargic just because of parasites might also alter the physiological as well as reproductive functions of hosts.

Table 1- Species diversity spectrum of endohelminths sampled from host-I *C.auratus*

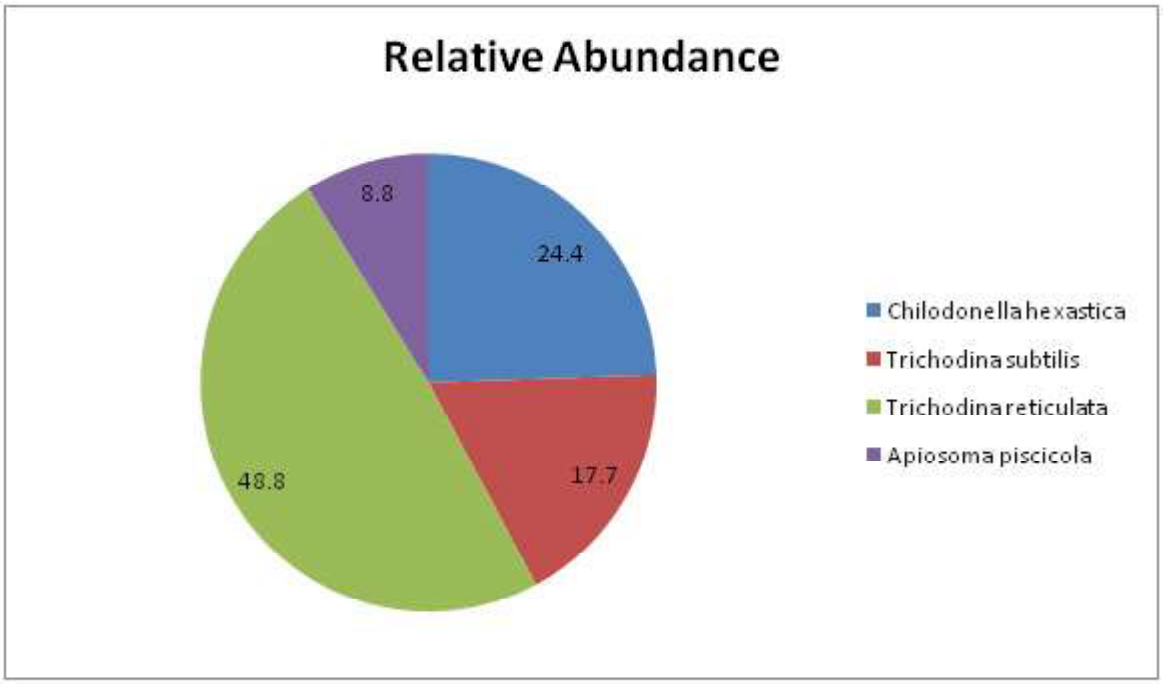
Fish host	Endohelminths recorded	No. of individuals sampled	*pi* (n / N)	Relative abundance RA= n/N x 100	*log pi*	*pi x log pi*	S-W Diversity, $= -\sum pi \, x \log pi$
Carassius auratus	*Chilodonella hexastica*	11	0.244	24.4	-0.612	-0.149	0.525
	Trichodina subtilis	08	0.177	17.7	-0.752	-0.133	
	Trichodina reticulata	22	0.488	48.8	-0.311	-0.151	
	Apiosoma piscicola	04	0.088	8.8	-1.055	-0.092	
	Total	**45**					

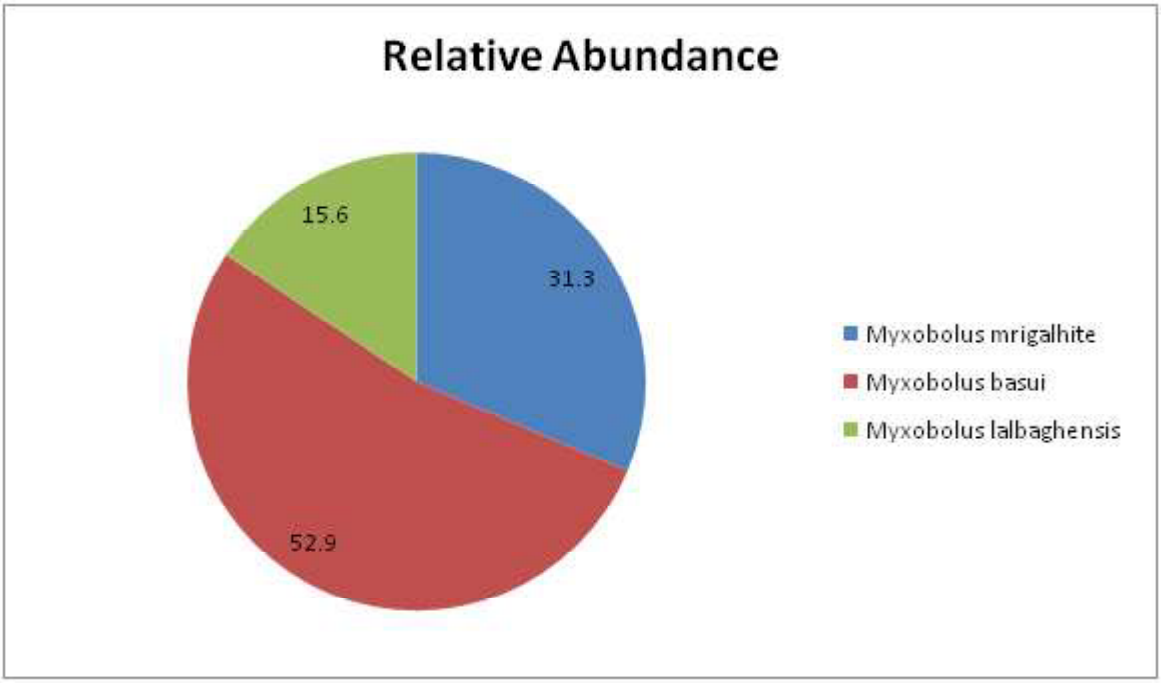
Table 2- Species diversity spectrum of endohelminths sampled from host-II *C.cirrhosus*

Fish host	Endohelminths recorded	No. of individuals sampled	*pi* (n / N)	Relative abundance RA= n/N x 100	*log pi*	*pi x log pi*	S-W Diversity, $= -\sum pi \, x \log pi$
Cirrhinus cirrhosus	*Myxobolus mrigalhite*	16	0.313	31.3	-0.504	-0.157	0.428
	Myxobolus basui	27	0.529	52.9	-0.276	-0.146	
	Myxobolus lalbaghensis	08	0.156	15.6	-0.806	-0.125	
	Total	**51**					

CONCLUSION

Monthly observations showed that the freshwater fishes from Madhepura district harbour a heavy burden of helminthic infection. Nematodes were the dominant parasites in *Carassius auratus* and *Cirrhinus cirrosus* fish species. They were mostly found in intestine and liver. The annotations of the present work could find out the species *Chilodonella hexastica, Trichodina subtilis, Trichodina reticulate, Apiosoma piscicola, Myxobolus mrigalhite, Myxobolus basui,* and *Myxobolus lalbaghensis*.


Relative Abundance
8.8
24.4
48.8
17.7
Chilodonella hexastica
Trichodina subtilis
Trichodina reticulata
Apiosoma piscicola


Relative Abundance
15.6
31.3
52.9
Myxobolus mrigalhite
Myxobolus basui
Myxobolus lalbaghensis

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Investigation of inhibition of protein denaturation assay by chitosan extracted from exoskeleton of *Sartoriana spinigera* (Wood-Mason, 1871)

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Abstract : Chitosan extracted from exoskeleton (carapace) of freshwater edible crab *Sartoriana spinigera* (Wood-Mason, 1871)[1] was assessed for *in-vitro* Protein (BSA) denaturation inhibition. The % Protein Denaturation Inhibitory activity of chitosan was tested at subsequent concentrations of 50 µg/ml, 100 µg/ml, 150 µg/ml, 200 µg/ml and 250 µg/ml and the present results exhibited a concentration dependent inhibition of protein (BSA) denaturation by the test extract i.e. 42.35±0.08, 49.41±0.9%, 58.82±0.67%, 69.41±0.60% and 83.53±1.03% respectively. Diclofenac Sodium was used as reference drug. 50 % Inhibitory Concentration (IC_{50}) values on protein (BSA) denaturation of Diclofenac sodium was 53.57(µg/ml) while IC_{50} value of the carapace chitosan was found to be 97.72 (µg/ml).

Keywords- *In vitro* Protein denaturation inhibition assay, *Sartoriana spinigera,* chitosan, IC_{50}

INTRODUCTION

Protein denaturation has been correlated with the formation of inflammatory disorders like rheumatoid arthritis, diabetes and cancer etc. Therefore, ability of substance to prevent the protein denaturation may also help to prevent the inflammatory disorders (Sangeetha G., Vidhya R., 2016)[2].

The most commonly used drug for management of inflammatory conditions are non-steroidal anti-inflammatory drugs (NSAIDs), which have several adverse effects especially gastric irritation leading to formation of gastric ulcers (Tripathi K.D., 2008)[3], (Bennett P.N. & Brown M.J., 2005)[4].

Natural resources have contributed significantly the development of modern medicine. Recently traditional folk medicine worldwide is being re-evaluated by extensive research on different plant and animal species and their active therapeutic principles. The rich wealth of flora and fauna can represent a novel source of newer compounds with significant anti-inflammatory activities. The major merits of traditional medicine seem to be their perceived efficacy, low incidence of serious adverse effects and low cost.

Although different parts of crabs are used in ethnomedicine for the treatment of inflammatory and related disorders by tribal folk in Jharkhand but the use of carapace chitosan extracted from *Sartoriana spinigera* and its anti-arthritic property have not been pharmacologically evaluated. Hence the present study undertaken to evaluate the anti-arthritic activity of carapace chitosan of *Sartoriana spinigera* by *in vitro* method.

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MATERIALS & METHODS

a) Collection of samples: Live specimens of freshwater crab *S. spinigera* were purchased from local market of Ranchi, Jharkhand and kept in aquarium in research laboratory of P.G. Department of Zoology, Ranchi University.

b) Extraction of chitosan: After 7 days acclimatization, crabs were narcotized and carapace were removed, washed, cleaned thoroughly and dried in hot air oven at 37°C. Then they were grinded to fine particles. Chitosan was extracted using the chemical process (Takiguchi, 1991)[5]. The demineralization process was done by using hydrochloric acid. The filtrate was soaked in 4% NaOH for 24 hrs at 80°C for deproteinization. The powdered form of Chitin obtained was deacetylated with 50% NaOH heated at 110°C and the filtrate was dried to obtain Chitosan.

c) Characterization of chitosan: - Samples of chitosan extracted from carapace of crab *S. spinigera* were sent to Central Instrumentation Facility, BIT Mesra for FTIR spectroscopy**.** Degree of deacetylation was calculated by graph obtained from FTIR-spectroscopy following the formula Brugnerotto, J. *et al*., (2001)[6]: **Degree of deacetylation (DD %)** = 100-Degree of acetylation. **Degree of acetylation** (%) = 31.92 × A1320 /A1420 – 12.20, where, A1320 and A1420 represents the absorbance at 1320 and 1420 cm^{-1} respectively.

d) Evaluation of *in vitro* anti-arthritic activity (R. Lavanya *et al.*, 2010)[7]:

♦ **Assay method:** In this model assessment of anti-arthritic activity of chitosan was studied by using inhibition of protein (BSA) denaturation technique which was done according to R. Lavanya *et al*., (2010)[7]. The test sample as well as standard drug solutions i.e. chitosan of carapace were taken in different concentrations- 50µg/ml, 100µg/ml, 150µg/ml, 200µg/ml and 250µg/ml and assayed in triplicates. The reaction mixture 0.5ml consisted of 0.45ml bovine serum albumin (BSA) of 5% aqueous solution + 0.05ml of 50-250µg/ml of extract. Distilled water instead of extracts with above mixture was used as a negative/ test control and positive/product control 0.5ml consisted distilled water and test solution. The pH of all the prepared solutions were adjusted to 6.3 with the help of 1N HCl. Diclofenac sodium was used as standard drug. All the solutions were incubated at 37^0 C for 20 minutes. After that, all the solutions were kept at 57^0 C for 3 minutes. And after cooling, 2.5 ml of 0.1 M phosphate buffer was added to the all test tubes. Absorbance was read at 416nm.

♦ Percent (%) inhibition of Protein (BSA) denaturation was calculated using the formula: -

[100 - {(OD of TEST SOLUTION - OD of PRODUCT CONTROL) / OD of TEST CONTROL} × 100]

STATISTICAL ANALYSIS:

All *in vitro* assays data signify the mean ± standard deviation of triplicates. The extract/drug concentration for 50% inhibition (IC_{50}) was determined form the dose response curve by plotting percentage inhibition with respect to control against treatment concentration.

RESULTS & DISCUSSIONS

FT-IR studies: The FT-IR spectrum of chitosan in the present study was obtained and effective peaks were compared with that of standard chitosan. The FT-IR spectrum of the standard chitosan showed nine major peaks at the ranges of 752, 893, 1022, 1421, 1552, 1643, 2865, 2921 and 3283 cm^{-1} (Puvvada *et al*., 2012)[8] peaks of chitosan sample extracted from *Sartoriana spinigera* were compared. Only 7 out of 9 peaks were nearly similar with that of standard 709.80, 875.68, 1026.13, 1489.76, 1658.78, 2885.51 and 3267.41 cm^{-1}. **DD (%):** Degree of deacetylation affects the chemical, physical and biological properties of chitosan such as adsorption, covalent linking, encapsulation (Puvvada *et al*. 2012)[8]. Degree of deacetylation (DD %) of the chitosan was found 98.19%, it is higher than reported by Puvvada *et al*. (2012)[8] in which degree of deacetylation of chitosan of shrimp waste was 89.79%. The obtained chitosan was found to be soluble in 1% acetic acid and the pH was found to be 8. FT-IR studies confirmed that the obtained product was chitosan.

Table 1: Percentage inhibition of protein (BSA) denaturation by the carapace chitosan sample and Diclofenac sodium

Concentrations (µg/ml)	% inhibition of Protein (BSA) Denaturation	
	Diclofenac sodium	Carapace Chitosan of *S. spinigera*
50	54.12±0.85	42.35±0.85
100	55.29±0.75	49.41±0.9
150	61.18±1.07	58.82±0.67
200	71.76±0.60	69.41±0.60
250	87.06±1.70	83.53±1.03

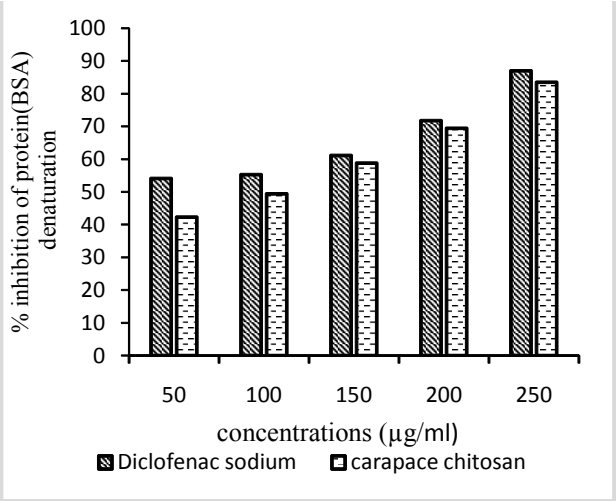


Fig 1: - % inhibition of protein (BSA) denaturation by Diclofenac sodium and carapace chitosan.

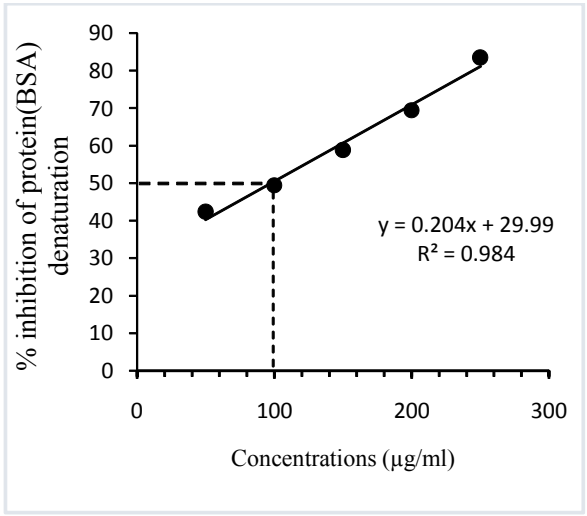


Fig 2: - 50% inhibitory concentration (IC_{50}) value for carapace chitosan was 97.72 (µg/ml)

Table 2: - 50 % inhibitory concentration (IC_{50}) values on protein (BSA) denaturation

Groups	IC_{50} (µg/ml)
Standard (Diclofenac sodium)	53.57
Test (Carapace chitosan)	97.72

The present investigation showed % inhibition of protein (BSA) denaturation by sample as well as Diclofenac sodium drug. Protein (BSA) denaturation inhibition assay was done with five different (50µg, 100µg, 150µg, 200µg and 250µg) concentrations of chitosan sample as well as standard drug. An *in-vitro* anti-inflammatory study revealed that Chitosan sample with 50µg showed 42.35±0.85% inhibition and it gradually increased (Fig. 1) to 49.41±0.9%, 58.82±0.67%, 69.41±0.60%, 83.53±1.03% inhibition for 100µg, 150µg, 200µg and 250µg respectively whereas in case of Diclofenac sodium at 50µg showed 54.12±0.85% and at 250µg showed 87.06±1.70 % inhibition. The results are summarized in Table (1), the present findings exhibited a concentration dependent inhibition of protein denaturation by the test extract throughout the concentration range of 50µg, 100µg, 150µg, 200µg and 250µg/ml. Diclofenac sodium (at the concentration range of 50µg, 100µg, 150µg, 200µg and 250µg/ml) was used as the reference drug which exhibited highest inhibition of protein denaturation. This was further confirmed by comparing their IC_{50} values. R. Lavanya *et al.* (2010)[7] reported that methanolic extracts of *Anisomeles malabarica* L. at a dose of 250µg/ml protein (BSA) denaturation was found to be 97.47%. While in this study the chitosan showed 83.53±1.03% at 250µg/ml. In this study the chitosan showed 83.53±1.03% at 250µg/ml concentration which is higher than the report of Darsan B. Menon *et al.* (2011)[9], they reported that the percentage inhibition of protein (BSA) by methanolic extract of the stem of *Plectranthus hadiensis* and standard Diclofenac, at 1mg/mL concentration, was 86.10% and 92.93% respectively. Srividya and Chandra (2015)[10] reported Inhibition of protein (albumin) denaturation at 200µg/ml was found to be highest in *Phyllanthus emblica* followed by *Limonia acidissima, Syzygium cumini, Anacardium occidentale, Carissa congesta, Artocarpus hirsutus* and the values were 79.01±3.07%, 65.75±2.53%, 51.57±2.83%, 33.71±1.10%, 28.73±2.53% and 28.00±1.94% respectively. Percentage inhibition value for the standard diclofenac sodium was found to be 92.64±2.30%. **IC_{50}: -** in this present study 50 % Inhibitory Concentration (IC_{50}) values on protein (BSA) denaturation of Diclofenac sodium was 53.57(µg/ml) while IC_{50} value of the carapace chitosan was found to be 97.72 (µg/ml) (Table-2), which is lesser than the *in vitro* protein denaturation assay of fresh water edible snail *Bellamya*

bengalensis reported by Anjan *et al.* (2015)[11] ,they found that the 50% Inhibitory Concentration (IC_{50}) of extrapallial fluid of the mollusca *Bellamya bengalensis* on protein denaturation was 87.56 µl/ml .

CONCLUSION

Carapace chitosan extracted from *Sartoriana spinigera* and its *in vitro* anti-arthritic property have not been pharmacologically evaluated. Thus, on the basis of the data obtained in this study it may be concluded that the chitosan extracted from carapace of *Sartoriana spinigera* contains the capacity to inhibit the protein denaturation *in vitro* and confirms its anti-inflammatory (anti-arthritic) properties. Further studies may require for its mechanism of action.

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Antibacterial activity of chitosan of freshwater crab, *Maydelliathelphusa masoniana* (Henderson, 1893) against *Staphylococcus aureus* and *Escherichia coli.*

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Abstract : Chitosan is known for its diverse functional properties due to its biological activities and application in the pharmaceutical and food industries. Antibacterial properties of chitosan extracted from shell waste of freshwater crab, *Maydelliathelphusa masoniana* of Jharkhand were determined against Gram negative bacteria, *Escherichia coli* (ATCC 25922) and Gram positive bacteria, *Staphylococcus aureus* (ATCC 25923). The antibacterial activity of chitosan (5mg/ml) in 0.2% acetic acid, was tested by Well-Diffusion method. Zone of inhibition measured for *E.coli* was 19.33±0.57 mm and 18.66±0.77 mm for *S.aureus*. Oflaxacin (5mcg) drug showed 31.66±0.57 mm for *S.aureus* and 29.66±0.57 mm for *E. coli*. Comparative statistical analysis by Student's t-test showed that the antibacterial efficacy of chitosan solution was similar both for *E.coli* and *S.aureus*. Negative control (0.2% acetic acid) does not show any zone of inhibition. These results indicate that chitosan from shell waste of freshwater crab *Maydelliathelphusa masoniana* could be used as an effective antibacterial agent against *E.coli* and *S.aureus.*

Keywords- *M.masoniana*, Antibacterial, Oflaxacin, *E.coli, S.aureus*

INTRODUCTION

Crab belongs to the largest distributed Phylum Arthropoda among invertebrates under the animal Kingdom. It is well known due to its consumption as food and also as medicine against many health problems. Generally people consume its meat and throw its shell as waste. In recent times, shell wastes are used as a source of biopolymer, Chitosan. Chitosan is mainly known for usage in food, pharmaceuticals and cosmetics industries Chitosan is non-toxic, biodegradable, polymer of D-glucosamine, linked by 1,4 -glycosidic bonds, obtained by deacetylation of Chitin, a polysaccharide of exoskeleton of freshwater as well as marine arthropods and molluscs.

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Traditional antimicrobials have been used as reliable preservatives to control microbial hazards in the food industry for decades[1]. But these widely accepted compounds are synthetic with harmful side effects. To minimize these effects, there is a need of natural and healthy source of medicines. Due to the negative impact from chemical preservatives, attention has shifted to the use of naturally derived antimicrobial agent to control food borne pathogen[2]. With the increasing claim for food safety and health standards, consumers have been more concerned about the occurrence of chemical residues in the food products[3], therefore natural antimicrobials are considered better than traditional synthetic antimicrobials. Natural antimicrobials are derived from many, including animals (chitosan).[4]

Chitosan have attracted the interest of many researchers, medical, pharmaceutical and industrial fields due to its properties like analgesic, antitumor, antioxidant, haemostatic, hypocholesterolemic, biodegradability and biocompatibility.[5] Chitosan was shown to have several advantages over other disinfectants, as it possesses a higher antibacterial activity, a broader spectrum of activity, a higher kill rate, and lower toxicity towards mammalian cells[6,7]. In the present study, Chitosan is extracted from the shell of one such locally found freshwater crab of Jharkhand*, Maydelliathelphusa masoniana* to determine its antibacterial activity.

MATERIALS & METHODS

- **Collection of animals**

Maydelliathelphusa masoniana were purchased from local market of Ranchi, Jharkhand. Shells were scraped free of loose tissue from the crab wastes in laboratory, washed thoroughly with tap water to remove impurities. They were dried at 60°C and pulverized using pestle and mortar for further analysis.

- **Preparation of chitosan and chitosan solution**

Chitin and chitosan were prepared from *Maydelliathelphusa masoniana* shell waste according to *Takiguchi*[8,9] with some modifications[10] for purification of chitosan. The production of chitosan from crustaceans shell generally consists of three basic steps demineralization, deproteinization and deacetylation. In the preparation of chitosan solutions 0.5% (w/v) chitosan were dispersed in 0.2% (v/v) acetic acid solution.

- **Bacterial Strains**

The antibacterial activity of the prepared chitosan from *Maydelliathelphusa masoniana* was tested against two strains, *Staphylococcus aureus* (ATCC 25922) and *Escherichia coli* (ATCC 25922), obtained from the Department of Microbiology, Rajendra Institute of Medical Science, Ranchi, Jharkhand. *Staphylococcus aureus,* a Gram positive bacterium and *Escherichia coli,* a Gram negative bacterium were chosen because both are commonly associated with food products and Urinary tract infections.

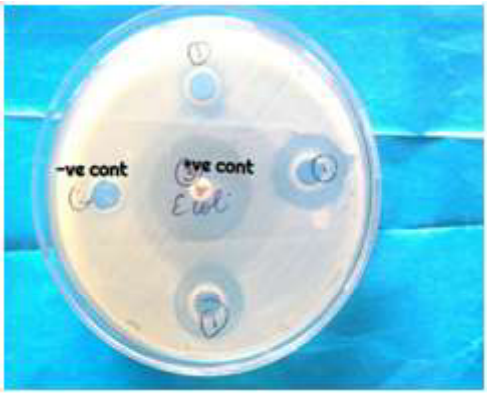
Nutrient broth was prepared and sterilized in an autoclave at 15 lbs pressure for 15 min. Individual species of bacteria were inoculated in the sterile nutrient broth and incubated at 37°C for 24 hrs. Muller Hilton Agar (MHA, Himedia) medium was prepared, sterilized in an autoclave at 15lbs pressure for 15 min and poured into sterile petridishes and incubated at 37°C for 24 hrs. The antibacterial activity of the individual bacterial strains was tested using Agar Well diffusion method[11]. Wells of 6mm diameter were made aseptically in the inoculated plates. Bacterial cultures were emulsified in normal saline and turbidity was matched with 0.5% McFarland turbidity standards. 24 hrs old nutrient broth cultures of test bacteria were aseptically swabbed on sterile MHA plates. Using micropipette, solution of chitosan (5mg/ml) in 0.2% acetic acid and 0.2% acetic acid as negative control was loaded in the respective wells. Ofloxacin disc (5mcg) was placed using a sterile forcep, as positive control. The plates were incubated at 37°C for 24 hrs in upright position. The antibacterial assay was carried out in triplicate. After incubation at 37°C for 24 hrs, zone of inhibition was measured in millimetres.

RESULTS

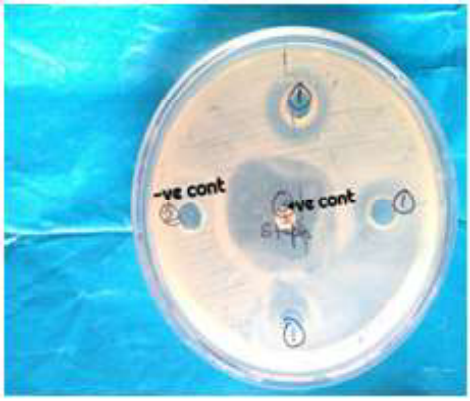
Table 1: Antibacterial activity and inhibition zone of chitosan (mm)

Sl. no.	Bacterial strains	Zone of inhibition (mm)				Negative control	t-test value
		I	II	III	Mean ± sd		
1.	*Staphylococcus aureus* (ATCC 25923)	19	18	18	18.66±0.70	---	25.4***
	Positive control	31	32	32	31.66±0.57		
2.	*Escherichia coli* (ATCC 25922)	20	19	19	19.33±0.57	---	21.9***
	Positive control	29	30	30	30.29±0.57		

(---) No zone of Inhibition,0.2% acetic acid - Negative control, Ofloxacin (5mcg)-Positive control,
*** $p<0.001$ or significant at 0.1%



Escherichia coli



Staphylococcus aureus

Figure 1: Antibacterial activity of chitosan from *Maydelliathelphusa masoniana*

In Table 1, chitosan solution (5mg/ml) showed zone of inhibition of 18.66±0.70 mm against *S.aureus* (ATCC 25923) and 19.33±0.57 mm against *E.coli* (ATCC25922) whereas with negative control (0.2% acetic acid) gave no inhibition zone. Positive control, Ofloxacin (5mcg) showed 31.66±0.57 mm and 29.66±0.57 mm against *S.aureus* and *E.coli* respectively. Statistical analysis revealed that there was no significant difference between zone of inhibition against *S.aures* and *E.coli* advocating that chitosan solution have similar antibacterial efficacy for both the strains.

DISCUSSION & CONCLUSION

According to Jean[12], Ueno[13], chitosan possesses antimicrobial activity against a number of Gram-negative and Gram-positive bacteria. Similar investigations were also made by Md. Monarul Islam[14] that with 1% chitosan solution (in 1% acetic acid) treated against *S.aureus* and *E.coli.* Zone of inhibition observed was 13mm and 10mm. respectively. Annaian Shanmugam[15] also examined antibacterial activity of 5% chitosan solution (0.2% acetic acid) and observed 11-15 mm inhibition zone against *S.aureus* and *E.coli.*

0.5% chitosan solution of the present study showed wider spectrum of antibacterial activity than reported zone of inhibition of 3mm against *S.aureus* and *E.coli* from chitosan (50mg/ml) in 0.1% acetic acid extracted from shell waste of *Portunus pelagicus* by P. Raja[16]. K.Prabu[18] reported that *S.aureus* showed inhibition zone of 8.17±1.21mm and 13.04±1.11 mm was formed against *E.coli* with chitosan solution (500mcg/ml) in 1% acetic acid, from another marine crab, *Podophalamus vigil.*

Whereas chitosan of the present study showed narrower spectrum of antibacterial activity than reported zone of inhibition of 55.4±0.71 mm and 49.7±0.31 mm against *E.coli* and *S.aureus* respectively from 5% chitosan in 0.1% acetic acid extracted from shell waste of crab obtained from Cario, Egypt.[2] A.Shanumugam[17] reported very good antibacterial activity against *E.coli* with zone of inhibition above 16mm diameter and good activity against with 11-15mm diameter against *S.aureus* by chitosan (5mg/ml) in 0.2% acetic acid extracted from cuttlebone of *Sepia kobiensis*.

In the present study, *S.aureus* and *E.coli* showed more sensitivity against chitosan than the previously mentioned reports. In the previous investigation, chitosan was extracted from cuttlebone of *Sepia kobiensis* and marine crab and shrimp but in present study chitosan was extracted from freshwater crab. Thus this can be the reason for better antibacterial efficacy of chitosan extracted from freshwater crab *Maydelliathelphusa masoniana.* In conclusion, the present investigation revealed that the chitosan from *Maydelliathelphusa masoniana* inhibits growth of human pathogenic bacterial strains, *S.aureus* and *E.coli.*

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Socio economic and aesthetic paradigms of lady bugs' diversity

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Abstract : Biodiversity in the recent years has been variously defined, understood and interpreted but seldom with holistic approach. The environment being a part of the great biosphere in combination with varieties of flora and fauna plays pivotal role in the formation of holistic concept of biodiversity. Socio economic and aesthetic approach of dealing biodiversity is one of the important contemporary dimensions which may further create the functional model or paradigms for understanding the environmental role of this component.

Biodiversity of predaceous coccinellids representing the lady bugs and their role as bioindicators in agro-ecosystem has pioneering role in the development of strategies for biological pest control having immense practical and scientific interest. The species of lady bugs- *Coccinella septumpunctata*, *Epilachna vigintioctopunctata* and *Rodolia amabilis* found in agro ecosystem in terms of biodiversity can be recruited as bioindicators due to their climatic and trophic characteristics. These beetles were abundantly found and sampled from two types of agro ecosystem of Ranchi-crop & forest area. The statistical computations of sampled data of the bugs have been found to be of significant value as bioindicator pronouncing the role in socioeconomic and aesthetic paradigms. These profiles of biodiversity are judging parameters of socio economic conditions of farmers along with the attractive and beautiful aesthetic scenario created by the colorful beetles wriggling on the leaves of the host plants.

Keywords- : Biodiversity, Agro forest, Lady Bugs, socio economic and aesthetic bearings.

INTRODUCTION

With the advancement of inter disciplinary knowledge in the field of environmental biology and biodiversity, a recent approach of socioeconomic and aesthetic value of biowealth has been developed in order to make the issues of biodiversity more useful and logical[1]. As a matter of fact, it is very hard to understand the fine fibres of environmental manifestations in which both abiotic and biotic components are beautifully integrated to make the environment a perfect natural laboratory of life science.[2]

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Socioeconomic consideration of any subject especially that of biodiversity of any group is directly concerned with the harmful and beneficial effects of the group of organism and their traditional practices adapted by both rural and urban human society.[3,4]

The urban folk are further making continuous research and design at provide an aesthetic fragrance to biodiversity with natural approach by promulgating conservational and eradicative strategies for the economically important group of flora and fauna.[5]

The lady bugs diversity particularly considered in the present research article has been evaluated through standard species diversity indices and the values have been

compared with the standard for interpreting the degree of successful survival and security of the species in the natural habitat of the bugs.

The successful and secured species of lady bug having diversity value more than 0.5 are generally considered to be more beneficial as biological control predator and species having less than 0.5 value are said to be self struggling species in order to establishment in the environment thereby discharging low degree of biopredation.[6]

The paper also deals with the aesthetic aspect of lady bugs' elytra coloration and number of spots present on them. The eternal linkage of colored spotted wings of these bugs with the environmental well being is not very clear so far except few astrologists trying to establish the relationship between the radiant photic energy coming out from the wings with the planetary position.[7]

However, in recent days designers of cloths, interior living space, exterior getup of the building and other structures have started using the bugs' colorful wing pattern, spots and combinations in their products to make it aesthetic and attractive.[8,9]

Apart from the routine study of biological characters, taxonomic descriptions and life cycle generation in the laboratory, most of the statistical analysis related with the diversity of the bugs' has been done on the field work survey and sampling of the beetles. Inferential concepts of socioeconomic and aesthetic paradigms of the bugs' diversity have been discussed with literary approach which may be imaginary but logical.[10]

MATERIALS AND METHODS

Species of Ladybug were randomly sampled during morning and evening hour from the agricultural field and residential garden as well as road side's weeds and herbs of Kanke, Ranchi during post rains seasons of 2018.

Sampling data were generated by numerical counting of the live bug captured during mating and no sacrifice of the living individual was made.

Keeping in view the restriction of biodiversity species and conservation of wild life. Individual farmer's enquiries were made to collect ethnoscientific information about the use of beetles/bugs as biological pest control (BPC) agent and predatory role of the three beetles being reported in this work.

All the field data regarding the population density, farmer's information and sampling photography have been statistically testified to establish the correlation between the bugs diversity and socio economic as well as aesthetic paradigm, presented in the observation section.

OBSERVATION

On comparing the S-W diversity value of the beetle and the information of farmers regarding the use of the beetle as biological control agent or predator of the crop pest, a very significant result has been procured regarding its socioeconomic paradigm.

The individual diversity value of *Coccinella septumpunctata* (Seven spotted beetle is -0.136, of *Epilachna vigintioctopunctata* is -0.158 and that of *Rodolia amabilis* is - 0.089 which reflect that the second species is more secured and dominant over the first and third but the cumulative diversity value is only 0.383 which is less than 0.5 of standard index thereby making all the species collectively secured and well protected as biological pest control agent.

From the individual enquiry of local farmers and also on inspection of seed crop from where the bug have been sampled it becomes apparent that species one *Coccinella septumpunctata* having population density of 35 in very strong predator and become the first choice of the farmer for BPC (biological pest control).

The second and third species, *Epilachna vigintioctopunctata* and *Rodolia amabilis* are relatively less predatory in nature as reflected in table. 1 and fig. 4. Hence, they are the third and second choice of farmers as BPC agent.

On comparing the data and information of field sampling, farmers enquiry and photography of bug infested host plant with the statistical computation value of data, it is further apparent that the value of S-W diversity both at individual species level and cumulative community level support the derivation made from table. 1 as the different ranking of the bugs for BPC agent.

The socio economic value of these beetles in terms of degree of predation from this study can be ranked as- *C. septumpunctata* (rank. 1) > *R. amabilis* (rank. 2) > *E. vigintioctopunctata* (rank. 3).

Table 1 - Socio economic value of lady bugs in terms of predation

Sl. No.	Species	Population density per unit area	Host plant	Predatory profile			
				Prey (Pest of plant)	Type	Degree of predation	Farmers choice for BPC
1.	*Coccinella septumpunctata*	35	Mustard Plant, Vegetables leaves, Wheat	Aphids, Thysanoptera, Aleyrodidae	Polygenetic	70% (Strong)	I
2.	*Epilachna vigintioctopunctata* (Hadda Beetle)	20	Leaves of grains, potatoes, Beans.	Aphid	Monogenetic	20% (Week)	III
3.	*Rodolia amabilis*	5		Hemiptera, Pseudococcidae	Polygenetic	40% (Average)	II

Graph 1 - Graphical representation of Socio economic value of lady bugs in terms of predation

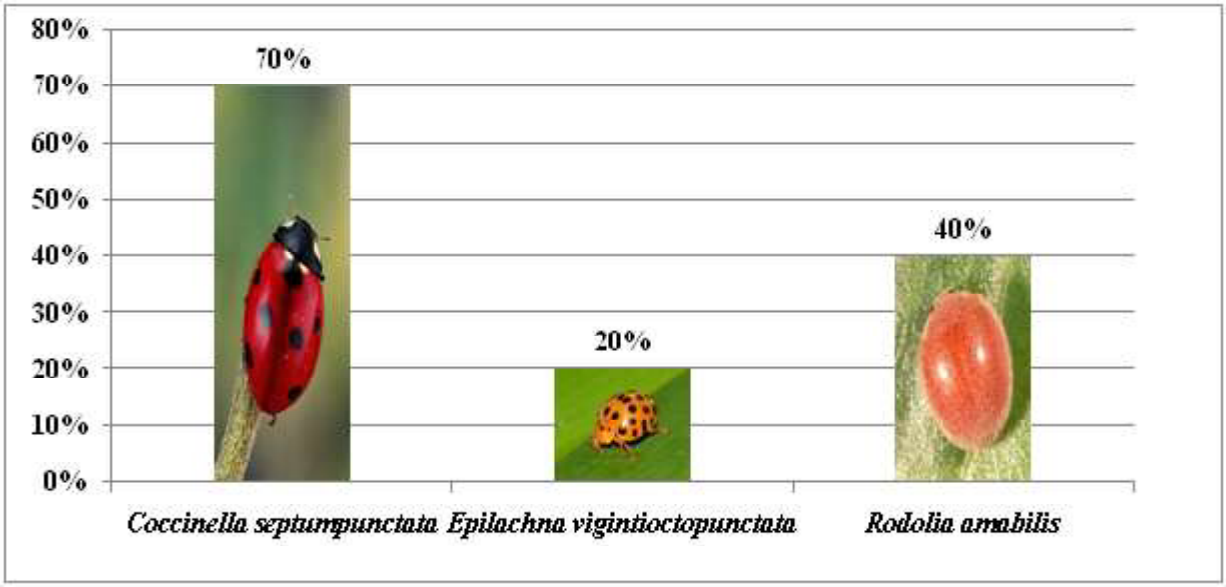


Table 2- S-W diversity of three ladybug species sampled from Kanke agriculture field, Ranchi

Sl. No.	Species	No. of Individuals sampled	*pi* (n / N)	*log pi*	*pi x log pi*	S-W Diversity, $H = -\sum pi \, x \log pi$
1	*Coccinella septumpunctata* (Seven spotted beetle)	35	0.583	-0.234	-0.136	0.383
2.	*Epilachna vigintioctopunctata* (Hadda Beetle)	20	0.333	-0.477	-0.158	
3.	*Rodolia amabilis*	5	0.083	-1.080	-0.089	
	TOTAL (N)	**60**				

Fig.1- Adult ladybeetle, *Coccinella septempunctata* (Seven spotted Beetle) and its larvae thriving on the tobacco leaves as well as voraciously predating on the aphids.



Adult of *C.septumpunctata*



Larva of *C.septumpunctata*

Fig. 2- Adult ladybeetle, *Epilachna vigintioctopunctata* (Hadda Beetle) and its larvae feeding on the potato leaves as well as scavenging the fungal layer as predator



Adult of *E.vigintioctopunctata*



Larva of *E.vigintioctopunctata*

Fig. 3 - Adult ladybeetle, *Rodolia amabilis* and its larvae thriving on the sugarcane leaves and predating on the hemipteran pests of sugarcane



Adult of *R.amabilis*



Larva of *R.amabilis*

CONCLUSION

3 species of lady bugs (*Coccinella septumpunctata, Epilachna vigintioctopunctata* & *Rodolia amabilis)* have been randomly sampled from different agricultural & vegetational fields as well as residential home garden of Kanke block, Ranchi, Jharkhand.

The average population density per unit quadrates range as below:-

C. septumpunctata (35) > *E. vigintioctopunctata* (20) > *R. amabilis* (5), this ranking of population density of bugs clearly denotes that *C. septumpunctata* is most dominant while *R. amabilis* is the recessive bug of sampling field.

Degree of predation of these bugs has been measured apparently in the range of 70%: 20%: 40% by ranked bug density (C.s> E.p>R.a). The S.W diversity value of these bugs randomly sampled and quadrateswise population data developed in the field also support that the most dominant predatory bug is *C. septumpunctata* and the most recessive is the *R.anabilis* (Ref. fig.)

The socio economic and aesthetic value of these bugs forming the diversity paradigm has been established by the statistical analysis of the population data sampled and observation of the biology- life cycle, type of metamorphosis, developmental stages, food plant consumption, rate of predation of the prey, infesting the host plant as pest and peculiar circle and line clustering of the bugs on the leaves of the host plant like nature worshiping for the proper rain agriculture and their produce as believed by the farmers, the tribal ethnic group & also mentioned in Atharva ved (seven spotted lady bugs clustering in group of seven individuals are natural host of seven colored rainbow inviting rain on the earth). The socio economic and aesthetic parameters of lady bug diversity paradigm are therefore integral component in which the statistical diversity indices and ethnic information of farmers have been clubbed.

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Studies on the effect of deltamethrin on the respiratory rate and some blood parameters of *Heteropneustes fossilis* (Bloch)

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Abstract : This study was to assess the effects of deltamethrin on some haematological parameters of *Heteropneustes fossilis* (Bloch). For 4 days, 30 *Heteropneustes fossilis* were exposed to two different concentrations of Deltamethrin which were 20% (0.91 ìg·L-1) and 40% (1.88 ìg·L-1) respectively. The results obtained from this study showed that the White Blood Cell (WBC), Haemoglobin content (Hb), Hematocrit percentage (PCV), Mean Corpuscular Volume (MCV) and Mean Corpuscular Hemoglobin (MCH) were decreased in treated groups when compared to the control ($p<0.05$). However, the number of Red Blood Cell (RBC) increased in deltamethrin treated fish ($p<0.05$). Mean Corpuscular Hemoglobin Concentration (MCHC) values were insignificant when compared to control group ($p>0.05$).

Key words: *Heteropneustes fossilis*, Deltamethrin, Effect, Hematological parameters, Toxicity.

INTRODUCTION

Deltamethrin is extensively used in agriculture, for controlling pests, insects and vectors of endemic diseases, protecting seeds during storage and fighting household insects because of their low environmental persistence.[1] The extensive use of deltamethrin on land were most probably washed into surface water and can adversely influence or kill the life of aquatic organisms and other higher animals. Aquatic organisms, particularly fish, are highly sensitive to deltamethrin. Due to their lipophilicity, pyrethroids have a high rate of gill absorption, which would be a contributing factor in the sensitivity of the fish to aqueous pyrethroid exposures. Fish seem to be deficient in the enzyme system that hydrolyzes pyrethroids[2]. It has been found that the fish exhibit several symptoms of stress when treated with deltamethrin[3]. Generally, deltamethrin affects the overall physiological profile in fish with particular reference to the energy metabolism, biochemical and haematological characteristics[4]. The mechanism of its neurotoxic effectiveness is the same as that of other type II pyrethroids: blockage of the sodium channels and inhibition of ã-aminobutyric acid receptors. Effects of deltamethrin on nervous, respiratory and haematological systems have been reported[5]. Possible toxicological actions of the pyrethroids include disruption of calcium and phosphate homeostasis and abnormalities in haematopoiesis and protein synthesis[6]. Deltamethrin causes severe morphological alterations in the gills and liver and causes toxic effects on the hematopoietic organs, liver and gills[7]. Deltamethrin may disturb the calcium and phosphate homeostasis and may lead to an effect on the reproductive state of the fish[8]. Many studies conducted in recent years have showed the harmful effects on fishes and the aquatic ecosystem of pesticides. Haematological parameters in fish living in the different environmental conditions are very

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important for toxicological studies[9-11]. Therefore, they are used as indicators of the physiological stress response to endogenous and exogenous changes in fish[12]. This study was to determinate the effects of deltamethrin on haematological parameters in *Heteropneustes fossilis*.

MATERIALS & METHODS

Collection and maintenance of fish: Fish samples were obtained from the local farm. The live *Heteropneustes fossilis* (n = 30) used in this study were mature, healthy, 1 years old and with an average weight of 150–190 g. The average water temperature was 10 ± 2°C during the tests. At the time of sample collection, fish had been fed with a commercial trout feed at 2% body weight once a day. Prior to the experiment, fish in each group were kept in 1 x 1.2m (wide-deep) fiber-glass tanks for 1 month. Aeration was provided along the experiments. The water quality parameters were measured as O_2 = 8.5 ppm, pH = 7.4, SO_4^{-2} = 0.30 mg/L, PO_4^{-3} = trace, NO_3^- = 3.15 mg/ L, NO_2^- = trace and conductivity = 230 ls/cm[12].

Acute toxicity: The effect of acute toxicity of the agricultural toxicant, deltamethrin was determined as LC50 (96h). For this purpose, 3 treatments including control were set up to test toxicity; each treatment was replicated three times with 10 fish per tank with 60 liters water capacity. Fishes were exposed to deltamethrin at a nominal concentration of 0.0 ìg·L^{-1} (control group), 0.91 (20%) ìg·L^{-1} and 1.88 (40%) ìg·L^{-1}, respectively, which were equivalent to approximately 30% and 60% of 96 h LC50 value (4.7ìg·L-1) for 4 days toxicity testing[11]. Mortality was recorded at 24, 48, 72 and 96 h after the beginning and dead fishes were removed immediately from the tank.

Haematology assay: After the test period, fish were quickly taken out from the water and held firmly on a bench with a cloth covering the head to obtain blood samples and blood samples were withdrawn from caudal vessels by a vacuum syringes and then samples were taken haematology tubes. The tubes filled with blood samples were shaken a few times in order to prevent freezing. Handling time of fish was less than 1 minute to minimize stress effects[13].

At the time of blood sampling, the appropriate smears were prepared for Giemsa staining. The smears were air-dried, fixed in 96% ethanol for 30 minutes and stained with Giemsa staining about 30 minutes. The smears were examined for leucocyte differential count under a compound microscope. The haematological parameters were examined for erythrocyte count (RBC, $\times 10^6$/mm^3), hematocrit (Hct, %), haemoglobin (Hb, g/dL), mean corpuscular haemoglobin (MCH= (Hb in gr / RBC in millions) × 10pg), mean corpuscular volume (MCV= (packed cell volume as percentage/RBC in millions) × 10ì3, fl), mean corpuscular haemoglobin concentration (MCHC= (Hb in g/packed cell volume) × 100 g per 100 mL, %), leucocyte count (WBC$\times 10^4$/mm^3) and differential leucocyte count[14].

Statistical analysis: The data obtained from this study was subjected to various statistical tools. The differences in the means (±SEM) between groups were assessed using Independent Samples-t test, adjusted to 95% confidence limits some important. SPSS program version 15.0 was used for statistical analysis.

RESULTS & DISCUSSION

Sub-lethal concentrations of deltamethrin were equivalent to approximately 20% and 40% of 96 h LC50 value for 4 day toxicity testing. In some cases, fish exposed to deltamethrin exhibited vertical and downward swimming, patterns, swimming near the water surface. Variations in haematology parameters such as RBC, WBC, PCV, Hb, MCH, MCV, MCHC levels were showed in Table 1. WBC, Hb and PCV levels were reduced after exposure to the pesticide deltamethrin. Nevertheless, RBC value increased significantly. Significant differences were observed in the RBC, WBC, PCV, Hb levels ($p<0.05$). The MCV and MCH values of exposed fish showed a significant decrease ($P<0.05$). There were slight fluctuations in the MCHC values. But, MCHC values were not significantly different from each other during the study ($P>0.05$).

Haematological parameters have been acknowledged as valuable tools for monitoring fish health, confirming maturation and monitoring any changes in the quality of water[15]. Therefore, a change in water is one of the most important factors for fish haematology. On the basis of haematological studies, it is done to determine the sensitivity to environmental conditions and toxic substances of fish[16].

The reduction of oxygen-carrying capacity in fish may be associated with a decrease in hemoglobin concentration that is affected by deltamethrin. The need for hemoglobin content increased. Hb content increasing

may depend on the deformed osmoregulation balance of fish. This chemical compound appears to interfere with the ability to bind hemoglobin to oxygen during respiration[17], It's reported that a significant decrease in Hb contents of *Oncorhynchus mykiss* and *Heteropneustes fossilis* which can be related with Hb destroying or a decrease in the rate of Hb synthesis. Both the decrease in hemoglobin concentration and the increase in the number of RBC can be explained with the increase in respiration of the fish for the oxygen they need. Similar changes in the white blood cell are also reported by Svobodova *et al.* (2003)[6] in carp following acute toxicity with permethrin and reported by Velisek *et al.* (2007)[18] in rainbow trout following acute toxicity with deltamethrin. Bradbury and Coast (1989)[19] reported signs of fenvalerate poisoning in fish.

CONCLUSION

Deltamethrin toxicity in *Heteropneustes fossilis* showed that it causes varied effects including oxidative stress, haematological changes. Also, 40% concentration of deltamethrin was found to be toxic to *Heteropneustes fossilis*. Further research is needed since different fish species and different growing stages of fish may respond to pollution factors at different levels.

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Taxonomic studies of Shell fish fauna of Madhepura, Bihar

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Abstract : In the present study the different species of shell fishes were considered for the taxonomic study in Madhepura. 15 species of shell fishes belonging to different class such as Gastropoda, Malacostraca and Bivalvia were studied. Among these classes of species the species belonging to the class Gastropoda were found to be 8 in number were dominant and outstanding shell fishes of the of Madhepura, North Bihar. Of Gastropoda, the species belonging to genera *Bellamya, Pila, Gyraulus, Indoplanorbis* were founded. Among Bivalvia species belonging to the genera *Parreysia, Lamellidens, Corbicula* were recognized but were only 4 in number. Finally among Malacostraca the species belonging to genera *Sartoriana, Penaeus* and *Macrobrachium* comprising of about 3 species. The region is biogeographically diversified to sustain a rich shell fish population which increases the ecological richness of the environment. Thus the result of the present study indicated that the Madhepura region of North Bihar is endowned with rich shell fish fauna.

Key words: *Heteropneustes fossilis*, Deltamethrin, Effect, Hematological parameters, Toxicity.

INTRODUCTION

Shellfish is a colloquial and fisheries term for exoskeleton-bearing aquatic invertebrates used as food, including various species of molluscs, crustaceans, and echinoderms. The term is often used for the edible species of the groups, especially those that are fished or raised commercially. Although most kinds of shellfish are harvested from saltwater environments, some are found in freshwater. In addition, a few species of land crabs are eaten, for example *Cardisoma guanhumi* in the Caribbean. Shellfish are among the most common food allergens. Despite the name, *shellfish* are not actually fish, but are simply water-dwelling animals. Shell fishes belong to Arthropoda and Mollusca which has been containing highest animal biodiversity of the biosphere. They form an important component of the food chain for the higher animal taxa transferring energy and matter from phytoplankton, macrophytes, and zooplankton to fishes, amphibians, reptiles, birds and mammals and other form of wild life. Despite the name shell fish are not actually sish but are simply water dwelling animals.

Considerable work has been done on the taxonomy, biology, ecology and behaviour of molluscs of Indian sub-continent (Hornell, 1921; Hora, 1925; Prasad, 1932; Tonapi, and Mulherkar, 1963; Tonapi 1971; Rao *et al.,* 1971; Subba Rao and Mitra, 1982; Sharma *et al.,* 1983; Singh, 1990; Roy, 2003)[1-10] but practically no information is available on the shell fisheries potential of the different areas of Madhepura district. The objective of this study

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was to know the taxonomic diversirty of shell fishes of wetlands of Madhepura.

MATERIALS AND METHODS

North-Bihar comprising districts of Madhepura was selected for the present study. Shell fishes exploited by different categories of people in these five districts were investigated. Extensive field trips were conducted regularly and data were collected after survey, interviews and on spot enquiries. The shell fishes were collected and preserved in 10% formalin and brought to the laboratory.

Each of the containers was labeled properly against the physical data sheet of sampling and brought to the laboratory for the identification and species confirmation. The intact animals were washed thoroughly in running tap water and slightly decalcified in aqueous acidic medium to find out growth rings. The specimens were identified with the help of available literature (Sharma *et al.*, 1983)[8]. The collected specimens were submitted in the museum of the University Department of Zoology, Madhepura.

STUDY AREA

North-Bihar situated at $83^0$21' to $88^0$17' NL and $24^0$55' to $27^0$31' EL has vast lentic resources in the form of ponds, tanks, ox-bow lakes, swamps, chaurs, canals, road and railway side depressions, flood plains and these water bodies have enormous potential for fish culture (Dehadrai, 1994; Verma, 1994)[11,12].

RESULT AND DISCUSSION

The following species were found during the study

1. Materials: *Bellamya bengalensis* race mandiensis (Kobelt)

Collection Site: Collected from the ponds and rivers of Haraili

Biology: Inhabit fresh water ponds, rivers, reservoirs, ditches and lakes and abundant in water having detritus, rich in macrovegetations on which they lay eggs specially on Eichhornia crassipes.

Taxonomic description

(i) Shell moderately thick globosely turbinate, broadly ovate and conic.

(ii) Whorls 4-5 regularly increasing convex with spaced spiral striae or bands, i.e. 5 in 4th whorl and 9th in body whorl and with transverse stripes of redish brown.

(iii) Foot is of moderate size and does not extend beyond the snout.

Niche: They were located on algal mat in abundance.

2. Materials: *Bellamya variata* (Viviparous) (Frauenfeld)

Collection site: Collected from canals and ponds of Madhepura districts of North Bihar.

Biology: Same as other *Viviparous* sp.

Taxonomic description

(i) It is smaller than other species of the genus.

(ii) Shell light to dark green in colour.

(iii) Whorls are 5 in number and moderately inflated or slightly convex.

(iv) Bands are absent.

(v) Body whorls are with well developed spiral keel at its middle level.

Niche: Located on decomposed materials of ponds and slow running water canals and covered with algae and adjacent materials of aquatic plants.

3. Materials: *Pila globosa* (Bolten)

Collection site: Collected from the paddy fields, Baiwah of Madhepura North Bihar

Biology: It inhabits freshwater ponds, lakes, rivers, tanks and quite abundant in water having succulent aquatic vegetations on which it feeds. It lives both in aquatic and terrestrial environments. During prolonged drought they may remain in diapause for a long time and after rains they return to normalcy.

4. Materials: *Paratelphusa spinigera*

Collection site: Collected from Budhma river in Madhepura.

Biology: It makes burrows on banks of rivers and ponds. These crabs come out for feeding from burrows but with the slightest disturbance swiftly return to their hole. It feeds not only aquatic plants but also different kinds of smallest animals.

Taxonomic description

(i) Dorso-ventrally compressed and consist of a large, broad cephalothorax and stumpy abdomen.

(ii) The carapace is fused with epistome at the side and nearly always in the middle.

(iii) Rostrum absent.

(iv) Antennules and eye stalks are contained in the sockets of carapace.

(v) Third maxillipedes are broad, flat, valve-like covering the mouth parts.

(vi) Five pairs of thoracic legs are well developed.

(vii) Abdomen is reduced and fixed under cephalothorax.

(viii) Pleopods are greatly reduced, present only two pairs in male and 4 pairs in female.

Niche: Live in the burrow or holes present in banks of rivers, ponds and lakes.

5. Materials: *Lamellidens corrianus* (Lea)

Collection site: Collected from freshwater ponds, rivers, and lakes of Madhepura.

Biology: Bottom dweller of fresh-water ponds, rivers and lakes, usually burrows in the mud at the bottom of ponds by large ventral foot. It does not go deep in the burrow because the posterior extremities of the valves remain exposed for the inhalent and exhalent respiratory water currents and omnivorous in nature.

Taxonomic description

(i) Two pseudo cardinals present on left and right valve, bears two parallel lamellar pseudo cardinals.

(ii) Shell is strongly in equilateral, transverse, large and thinner.

(iii) Left valve is with single thin lamella-like cardinal teeth and the right valve is with two elongated, straight lateral teeth.

(iv) Umbos are slightly elevated and bear coarse ridges.

Niche: Located in mud of the ponds and were surrounded by different kinds of aquatic vegetations.

6. Materials: *Lamellidens marginalis* (Lamarck)

Collection site: Collected from the Loram river of Madhepura

Biology: Same as *Lamellidens corrianus*.

Taxonomix description

(i) Shell is transversely oblong, ovate or oval.

(ii) The umbos are swollen, knoblike placed much near the anterior end of each shell valve and bears coarse ridges.

Niche: It remains 1.5" deep inside the bottom.

7. Materials: *Gyraulus convexiusculus* (Hutton)

Collection: Collected from the ditches present besides river of Duas.

Biology: This snail is found in most of water bodies specially where there is plenty of aquatic weeds and plants. It is entirely herbivorous.

Taxonomic description

(i) It has thin transparent shell with very fine transverse striations and spiral ridge marking the shell.

(ii) The shell is small and discoidal with the whorls rounded to carinate.

(iii) The aperture is oblique and somewhat deflected.

Niche: It is located on aquatic weeds and macro vegetations of ditches.

8. Materials: *Indoplanorbis exustus* (Deshayes)

Collection site: Lacustrine habitats.

Biology: It is most common freshwater air-breathing gastropod found in grass- lands, ponds and lakes and herbivorous in habits.

Taxonomic description

(i) It is enclosed in brown, stout and spirally coiled shell.

(ii) Shell is sinistral and discoidal having a depressed spire.

(iii) Whorls of spire are 3 in numbers with transverse striations.

Niche: It is located on herbaceous grasses found besides rivers and ponds.

9. Materials: *Corbicula striatella* (Deshayes)

Collection site: Collected from the Koshi rivers of Khagaria

Biology: Inhabits in rivers and streams, where the sand or graval at the bottom act as a substratum. It is omnivorous in nature.

Taxonomic description

(i) Dark brown and strongly striated shell.

(ii) The shape is triangular and outer surface usually sculptured with concentric ridges and covered with greenish and shining peristracum.

(iii) There are two cardinal teeth in each valve, lateral teeth serrated.

(iv) Lines of growth well marked.

Niche: Located near bottom on substrate free grom rooted green algae.

10. Materials: *Melanoides tuberculatus* (Muller)

Collection site: same as above.

Biology: Same as *Melanoides lineatus*.

Taxonomic description

(i) The shell is highly elongated with many transverse ridges.

(ii) The average shell length is about 20–27 mm Glöer P. (2002)[13]. or 30–36 mm, but exceptional specimens may be up to 80 mm long.

(iii) The whorls of spire is convex and 8-9 in number.

(iv) Spire is strongly depressed.

(iv) Body whorl small and oval.

(v) Radula has a less broad and denticulate central tooth.

Niche: They were located from the bottom of ponds and were surrounded by different kinds of aquatic vegetations.

11. Materials: *Achatina (Lissachatina) fulica* (Bowdich)

Collection Site: Collected from gardens and grasses of river bank of Koshi.

Biology: It is primarly terrestrial pulmonate inhabiting open grasslands. It prefers humid area, specially the bank vegetations. They are mostly nocturnal and spend the day time under plants and aestivate during summer. They are herbivorous and eat the vegetations available in the area.

Taxonomic description

(i) Shell is elongated, thick and light horny colour.

(ii) Collumella and whorl of the shell is whitish with brown bands in adults.

(iii) Whorls of spire are 6 in numbers in which body whorl is oval and elongate.

(iv) Operculum absent.

(v) Foot is flat and modified for creeping by the large ventral sole.

(vi) The foot is provided with a large tubular slime gland, which produce profuse quantity of slime.

Niche: They were located on the garden plants including small to large vegetations.

12. Materials: *Pila virens* (Lamarck)

Collection site: Same as *Achatina (Lissachatina) fulica*

Biology: Same as *Pila globosa*

Taxonomic description

(i) Shell smaller than *Pila globosa*.

(ii) Spine more strongly depressed about ¼th of total length of shell

(iii) Whorls of the spire less marked and inflated.

(iv) Umbilicus is smaller and narrowly perforated.

(v) Operculum is horny, pale, fleshy brown and concave externally.

Niche: Present abundantly on detritus rich ponds, canals and paddy fields and were surrounded by carrions and humus.

13. Materials: *Sartoriana spinigera*

Collection site: Same as *Achatina (Lissachatina) fulica*

Biology: Same as *Pila globosa*

Taxonomic description

(i) *S. spinigera* possess brownish yellow, orange, brown or dark brown carapace colour and orange to brown chillete colour.

(ii) The carapace texture of the crab is very much oily.

(iii) A large 'V'shaped patter with dark brown spot present on the carapace of the crab. The crab is commonly known as Wood Meson crab.

(iv) These crabs are found crawling, burrowing and even buried in the mud soil of the littoral region of wetlands.

(v) The carapace of the crab is oily and smooth in texture; semi-triangular in shape which has many spines often camouflaged with algae and other encrusting species.

Niche: Present abundantly on detritus rich ponds, canals and paddy fields .

14. Materials: *Macrobrachium rosenbergii*

Collection site: Same as *Achatina (Lissachatina) fulica*

Biology: Same as *Pila globosa*

Taxonomic description

(i) *M. rosenbergii* is a striking looking prawn in which the second pair of walking legs can really justify the genus name meaning 'large arms'.

(ii) In the males these walking legs can have a vibrant shade of blue and can also be twice the body length.

(iii) The largest males can attain a total length from tip of rostrum to the end of the telson of 320 mm compared to 250 mm for the largest females (Holthuis, 1980)."[14]

(iv) The rostrum at the front end of the cephalothorax is very prominent with 11-14 dorsal teeth and 8-10 ventral teeth.

(v) The abdomen of the male is narrower and the female, as well as having a wider abdomen, has longer pleura.

Niche: found in rivers, ponds by river edges, up vertical surfaces (small waterfalls, weirs, etc.) and across land.

15. Materials: *Penaeus monodon*,(giant tiger prawn or Asian tiger shrimp, *Tresaugue*)

Collection site: Same as *Achatina* (Lissachatina) *fulica*

Biology: Same as *Pila globosa*

Taxonomic description

(i) Females can reach about 33 cm (13 in) long, but are typically 25–30 cm (10–12 in) long and weigh 200–320 g (7–11 oz); males are slightly smaller at 20–25 cm (8–10 in) long and weighing 100–170 g (3.5–6.0 oz).

(ii) The carapace and abdomen are transversely banded with alternative red and white.

(iii) The antennae are grayish brown.

(iv) Brown pereiopods and pleopods are present with fringing setae in red. Motoh, H (1981).[15]

Niche: Present abundantly on detritus rich ponds, canals.

DISCUSSION

The present study on the status of shell fisheries inhabitants of the area of Madhepura of North-Bihar reveals that the river basin of this region is rich in diversity of molluscan and crustacean fauna. The diversity of shell fishes in this region indicates well established balanced ecosystem. The abundance of shell fishes in terms of taxonomic diversity indicateds a good life support system for fishes and birds. Shell fishes are the major component of the macro-invertebrates, they form link between zooplankton and vertebrate taxa, such as fishes and birds and play a key role in the energy flow and bio-geochemical cycle of the wetland habitats.

Total number of species collected were about 15 from the different parts of Madhepura. Among these 15 species 8 belonged to the class Gastropoda, 4 belonged to the class Bivalvia and 3 belonged to the class Malacostraca. The taxonomic diversity in terms of number and abundance is more in Gastropoda and dominant species and were also among the edible aquatic animals while species belonging to Malacostraca were found to be least. It was observed that the appreciable seasonal changes of their population may be correlated with the appearance and disappearance of macrovegetations of the habitat. The emergent portion of the plants affords shelter to the adult stages of Mollusca. The floating vegetations constitute a biotope with varied ecological niches for almost all types of shell fishes found. Most of the molluscans were also found among the submerged parts of plants such as Gastropods. The free-floating vegetations such as *Eichhornia crassipes* afford the colonization of molluscs such *Bellamya bengalensis*, *Lamellidens marginalis, Corbicula* and others. Thus, macrophytes provide shade, shelter and site for oviposition and development of these shell fishes. A number of fish and avian fauna diversity directly depend upon the molluscs population of these habitats. The commercial aspect of shell fishes as raw material for food, finance, recreation, medicines, vitamins and minerals supplements etc., for local human population and ecological aspect for increasing biological diversity and maintaining ecological balance for the animals occupying the higher trophic level of the food chain. Thus, the considerable scope with respect to the shell fisheries for their medicinal value and these resources need judicious utilization on commercial basis to generate employment opportunity and enhance the income of the local people (Rao 1969).[16]

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Incidence of whiteflies pest (*Bemisia tabaci*) & degree of damage in the sunflower plants of Madhepura district

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Abstract : *Bemisia tabaci* (Gennadius) (Hemiptera: Aleyrodidae) is a pest species complex that causes widespread damage to cassava, a staple food crop for millions of households in India as well as other parts our country. Species in the complex cause direct feeding damage to cassava and are the vectors of multiple plant viruses. Here in this research work the focus has been given to study about the presence and damaged caused by the whiteflies on sunflower. Sun flower (*Helianthus annuus* L.) is one of the important oil seed crops and potentially fit in agricultural system and oil production sector of India. Various diseases, insects and nematodes attack damage the sunflower crop, results a wide range of loss in production and yield. Sunflower is susceptible to diseases of various kinds. The most serious diseases of sunflower are caused by fungi. The major diseases include rust, downy mildew, Verticillium wilt, Sclerotinia stalk and head rot, charcoal rot, blight and leaf spot.

Key words: *Bemisia tabaci*, virus, staple food, diseases.

INTRODUCTION

Insects are our rivals, because they are herbivore and ravaging our crops and are major threat to crop production. Some of them serve as vector for virus and other diseases. Leaf curling, chlorosis and premature senescence of plants can be due to severe infestation on sunflower by diseases, insects and nematodes that results a wide range of loss in production and yield of sunflower crop (Mukhtar, 2009).[1] Sattar *et al.,* (1984)[2] reported the insect pests of sunflower as whitefly, *Bemisia tabaci*; aphids, *Aphis gossypii*; jassids, *Amrasca devastans;* bud moth, *Heliothis armigera* and surface grass hopper, *Chrotogonus* spp. Kakakhel *et al*., (2000)[3] reported the insect pests attack on sunflower are *Bemisia tabaci*, *Empoasca* spp., *Thysanoplusia orichalce, Diacretia obliqua*, *Nezara viridula*, *Helicoverpa armigera* and *Nysius inconspicuus*.

Bemisia tabaci (Gennadius) (Hemiptera: Aleyrodidae) is a pest species complex that causes widespread damage to cassava, a staple food crop in many millions of smallholder households in India. *Bemisia tabaci* causes direct feeding damage to cassava, excretes sugar-rich honeydew, which acts as a substrate for sooty moulds that reduces both respiration and photosynthesis (Nelson, 2008)[4]. In addition, *B. tabaci* vector multiples plant viruses that cause two damaging diseases: cassava mosaic disease (CMD) and cassava brown streak disease (CBSD), that in combination lead to significant yield loss in cassava (Holt &Colvin,2001;Maruthi *et al.,*2002)[5,6]. This disproportionate

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approach to managing insect vectored plant diseases is not unusual, but has led repeatedly to management solutions that are not sustainable. This review of the empirical evidence is timely and necessary as we need to identify clearly the biotic and abiotic factors that may have contributed to high population growth of *B. tabaci* in the past, before we can develop urgently needed and sustainable management recommendations for the future.

Sunflower, *Helianthus annuus* (L.) popularly known as "Surajmukhi" belongs to the family Asteraceae and genus Helianthus is one of the most important oil crops globally and is grown on over 22 million hectares worldwide, with a production of 26 million tones . The name has its origin in Greek "Helios" means "Sun" and "Anthois" means "flower". Sunflower is an important oilseed crop in India. Sunflower originated in Southern United States and Mexico from where it was introduced into Europe and later into former Union of Soviet Socialist Republics (USSR). In India, sunflower crop was introduced in 1969. It is cultivated for edible seed (Khan *et al*., 2007)[7]. The agro-climatic condition of Pakistan is not conductive in general for palm oil cultivation. Sunflower and Soyabean were found to have the potential to raise the productivity of edible oil in Pakistan (Nasir, 2003)[8]. Over the years, sunflower has become an important crop for both farmers and consumers in Pakistan. Sunflower is a crop that fits well in the local cropping system and is considered the most important cash crop in all parts of the country.

MATERIAL & METHODS

Selection of the sampling site as follows up schedule of our study was done randomly and the sites were located in the northern zone of Madhepura district. Madhepura is located at 25°34'-26°07' North Latitude & 86°19'-87° 07'East Longitude. Madhepura's nearest district and surrounded by Araria & Supaul in the North, Khagaria & Bhagalpur in the South, Purnia district in the East and Saharsa in the West. It is situated in the plains of river Koshi & located in the Northeastern part of Bihar. The primary occupation of the people of the district is agriculture. The total geographical area of the district is 1788 Sq. Km.

The field experiment was laid in randomized block design during 2018 in four different areas of Madhepura where the production was very high. The four different areas that was chosen for the observation of incidence as well as damaged caused by the whiteflies were Gwalpara, Puraini, Bihariganj and Shankarpur (Fig: 1). These places were visited weekly after 15 days of germination to record the insect infestation till the maturity of the plant of sunflower. Observations on whitefly were made on leaves comprising of leaves each top, middle and bottom portion of randomly selected plants from each plot. The mean population of whitefly was worked out and subjected to statistical analysis.

Data Recording

Five plants were randomly selected for data recording and data regarding insect infestation was recorded weekly on per plant basis; (upper, middle and lower leaves of plants were examined) in case of whitefly larvae of Head moth were counted when founded in the heads of sunflower plants already selected.

RESULT & DISCUSSION

Incidence pattern and damage caused by whitefly (*B. tabaci*) on sunflower during 2017-18:-

There are many factors which affect production of the sunflower crop. Insects are major ones of it. About 43 insect species have been reported to attack the sunflower crop in India (Sandhu *et al.*, 1973)[9]. This finding is also in line with (Makhdoomi *et al*., 1984)[10] who also reported that 43 insect species attack this crop whereas (Hassan *et al*., 1984)[11] identified 19 insect species for the same. There is an evidence of insects which invade the sunflower crop plants significantly. *Helicaverpa armigera*, *Aphis gossypii* (Glover), *Bemisia tabaci* (Gennadius), *Amrasca devastans* (Ishida) and *Atractomorpha crenulata* (Fabricius) are reported to attack the sunflower crop (Sattar *et al*., 1984)[12].

In the present studies the damaged caused by the white flies was observed on sunflower in the four different areas of northern parts of Madhepura during the month of April, May and June. During the research period it was found that the Puraini was highly infested by the whiteflies (Table 3). While Shankarpur showed the lowest infestation of whiteflies.

The *Bemisia tabaci* is a polyphagous insect pest feeding on a number of economically important crops. Its infestation on sunflower crop has been reported in Israel

India (Sethi *et al*., 1978)[13] and Pakistan (Aslam *et al*., 2000; Kakakhel *et al*., 2000)[14,3]. In the present study, *B.tabaci* infestation occurred throughout the crop season and its population continuously increased until harvest of the crop. The population fluctuation of whitefly was monitored from third week of January to fourth week of May in three different dates of sowing (Table 2). In first date of sowing presence of whitefly on the crop was for eight weeks, which on second sowing was seven weeks at both sites. Whereas, on third date of sowing population were seven and six weeks respectively. It had been found that the plant harboured the pest only at vegetative stage and the attack had always been found to cease before capitulum formation stage. This feeding weakens plants, causing leaves to turn yellow, look dry or fall off plants early. A severe infestation can cause a loss of vigor, stunt plant growth or cause wilting.

Fig: 1- Different areas of Madhepura where the observation of whiteflies (*Bemisia tabaci*) was conducted



Table 1: Average insect pest (per plant) infestation on sunflower crop during April, May & June in Gwalpara

Sl .No	Date	Infestation
1	1/4/18	1.30±0.08
2	8/4/18	1.50±0.09
3	15/4/18	1.55±0.07
4	22/4/18	2.90±0.07
5	29/4/18	3.70±0.08
6	6/5/18	3.10±0.03
7	13/5/18	2.80±0.10
8	20/5/18	2.40±0.14
9	27/5/18	3.00±0.03
10	3/6/18	1.00±0.04
11	10/6/18	2.33±0.07
12	17/6/18	2.05±0.08
13	24/6/18	2.00±0.09

Table 2: Average insect pest (per plant) infestation on sunflower crop during April, May June in Puraini

Sl. No	Date	Infestation
1	1/4/18	1.10±0.15
2	8/4/18	2.50±0.09
3	15/4/18	1.55±0.07
4	22/4/18	2.90±0.07
5	29/4/18	3.70±0.08
6	6/5/18	3.80±0.03
7	13/5/18	2.80±0.10
8	20/5/18	3.10±0.20
9	27/5/18	2.36±0.40
10	3/6/18	3.00±0.10
11	10/6/18	2.18±0.32
12	17/6/18	1.50±0.50
13	24/6/18	1.00±0.30

Table 3: Average insect pest (per plant) infestation on sunflower crop during April, May June in Bihariganj

Sl. No	Date	Infestation
1	1/4/18	1.00±0.07
2	8/4/18	1.40±0.09
3	15/4/18	1.35±0.07
4	22/4/18	2.50±0.06
5	29/4/18	1.70±0.08
6	6/5/18	2.60±0.02
7	13/5/18	2.20±0.08
8	20/5/18	1.80±0.04
9	27/5/18	3.00±0.04
10	3/6/18	2.25±0.05
11	10/6/18	2.30±0.08
12	17/6/18	1.80±0.06
13	24/6/18	1.00±0.04

Table 4: Average insect pest (per plant) infestation on sunflower crop during April, May June in Shankarpara

Sl. No	Date	Infestation
1	7/4/18	1.00±0.01
2	14/4/18	1.30±0.05
3	21/4/18	1.05±0.03
4	28/4/18	1.50±0.02
5	5/5/18	2.00±0.04
6	12/5/18	2.10±0.06
7	19/5/18	1.40±0.09
8	26/5/18	2.20±0.07
9	2/6/18	2.10±0.02
10	9/6/18	1.00±0.01
11	16/6/18	1.50±0.05
12	23/6/18	1.40±0.08
13	30/6/18	1.00±0.01

CONCLUSION

Although Jassid/Leaf Hopper (*Amrasca biguttula*) and Head moth (*Helicoverpa armigera*) are the most notorious pests of the sunflower crop, whitefly (*Bemisia tabaci)* are also considered as one of the major pest of sunflower that led to decrease in its production. In this investigation the area that was highly infested by the whiteflies was Puraini area of Madhepura, while Shankarpara had the lowest infestation caused by the whiteflies. Screened sunflower hybrids showed significant susceptibility to the whiteflies (*Bemisia tabaci*) which were responsible for the loss in yield and for huge expenditures met by the farmers concerned in order to adopt measures to control them.

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Impact of water containing high arsenic on the blood metabolites in female mice.

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Received : 18th October 2018; Revised : 5th January 2019

Abstract : Arsenic is a naturally occurring toxic metalloid. Levels of naturally occurring arsenic in groundwater above the drinking water standard of 10 ìg /L are not uncommon. Inorganic arsenic (iAs) is a complete transplacental carcinogen in mice. Previous studies have demonstrated that *in utero* exposure to inorganic arsenic promotes cancer in adult mouse offspring, possibly acting through epigenetic mechanisms. Humans and rodents have the ability to convert arsenic to its methylated metabolites. Arsenic, from both natural and anthropogenic sources, is mainly transported in the environment by water. The form and concentration of arsenic depends on several factors, including whether the water is oxygenated, the degree of biological activity, the type of water source and the proximity of the water source to arsenic-rich geological formations and other anthropogenic sources.

Key words: Arsenic, blood metabolites, water, rodents.

INTRODUCTION

Arsenic is known as human carcinogen, inducing tumors of the skin, urinary bladder and lung. Arsenic, which is commonly found in drinking water, is a potent toxicant. Arsenic's modes of action include enzyme inhibition and oxidative stress as well as immune, endocrine, and epigenetic effects. Susceptibility to arsenic is dependent on the biomethylation, which occurs via one-carbon metabolism. Arsenic easily passes the placenta. The fetus and infant are probably partly protected by the increased methylation of arsenic during pregnancy and lactation; the infant is also protected by low arsenic excretion in breast milk. Early-life exposure may induce changes that will become apparent much later in life.

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Arsenic is increasingly found in the districts of Bihar, West Bengal, Uttar Pradesh and even Assam. According to World Health Organization, the permissible limit of arsenic in drinking water is 0.01 mg/l, which is equivalent to 10 ppb. Recently, however, it has been reported that there is an increased risk of arsenic toxicity, even at low and permissible dose of 10 ppb (Karagas *et al.* 2002)[1]. Arsenic exposure causes both acute and chronic toxicity in mice. The idea that arsenical induced toxicity could be modified by nutrients was initially proposed in the early 1930's by Mayer and Sulzberger (1931)[2], who suggested that adequate levels of ascorbic acid, in the diet prevented or reduced occurrence of arsenic induced anaphylaxis.

MATERIALS AND METHODS

Collection of Specimens

Female albino mice (27.73 ± 0.65g) were procured from the animal house of L.N.M.S. College, Birpur, Supaul.

They were housed in groups of five in polypropylene cages in an air conditioned room maintained at 25 ± 2°C, in a 12 hr light and 1 hr dark cycles. They were feed on bread, carrot and tap water.

Treatments

Twenty healthy female mice weighing 27.73 ± 0.65 g were elected for the present study. After acclimatization to laboratory, mice were divided into 4 groups, each containing 5 mice. Weights of mice were taken individually. Mice of group A were given normal food and drinking water and treated as controls.

Mice of group B were administered predetermined sub lethal dose of arsenic (3 mg/Kg body weight/day) through gavages for 15 days. This dose is well within the range for the human lethal dose (1-4 mg/kg body wt) reported for arsenic (North *et al.* 1997). Mice of group C were administered the same concentration of arsenic similar to group B.

Moreover they were administered with 10 mg/Kg body weight of ascorbic acid simultaneously every day for 15 days. This dose of ascorbic acid is quantitatively equivalent to the human therapeutic dose i.e. 500 mg/day (Sahu and Das 1994) in terms of body weight. Mice of group D were administered the dose of arsenic in same concentration and manner similar to the mice of group B and C. Moreover they were administered with 100 mg/Kg body weight garlic extract/mice simultaneously every day for 15 days.

The aqueous extract of garlic and ascorbic acid were given ½ hour prior to the arsenic dose. On the 16th day mice were weighed and sacrificed after giving light anesthesia. Weight of kidneys, liver and uterus were taken. Blood was collected through cardiac puncture and sent to a laboratory for analysis. Following parameters were analyzed: total WBC, differential leucocyte, total RBC, platelet counts, haemoglobin content, Serum Glutamic Oxaloacetic Transaminase (SGOT), Serum Glutamic Pyruvic Transaminase (SGPT), Serum Creatinine, Total protein, Blood Urea Nitrogen content and Blood sugar level.

Blood collection

At the end of the experimental period (30 days), animals were sacrificed by cervical dislocation. Immediately after that, a heart puncture was made and 2ml of blood was quickly collected " 1ml in 10% ethylene diamine tetra acetic acid (EDTA) tubes for the separation of plasma and 1ml in plain vacutainer, used for the assessment of different inflammatory markers. The blood collected from another set of mice was separated into serum and plasma and these had been used for the assessment of lipid profile level and hematological examinations, respectively.

OBSERVATION

Treatment	Control	Body weight
2 weeks	0 lg/ml	19.2 ± 0.2
	100 lg/ml	15.7 ± 0.4
10 weeks	0 lg/ml	23.4 ± 0.4
	100 lg/ml	20.8 ± 0.1

RESULT

The study concluded that low concentration of arsenic trioxide may have no significant increase and reduction in total body weight. Treatment with 100 lg/ml AsV (416 lg/ml sodium arsenate) in the drinking water had no effect on body weight gain in female or male mice (Table 4). However, female mice treated for 2 or 10 weeks with 100 lg/ml AsIII (173 lg/ml sodium arsenite) in the drinking water had significantly decreased body weights compared with their respective control groups.

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Studies on the effect of plant alkaloid-colchicine on the Gastro-intestinal physiology of female albino rat

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Abstract : Colchicine an alkaloid product of meadow-saffron plant *Colchicum autumnale* is powerful inhibitor of somatic cell multiplication through mitosis which may have distant bearing on the physiological function of the system. The female subjects compared to the males are supossed to be more suseptable to such alkaloids possibly due to lack of inhibiting cellular receptors for colchicine. In the present study different doses of colchicine administered in the female rat have been found to influence the intestinal function significantly leading to rapid breakdown of the mucosa cells. Correlation to this the rate of cellular absorption of sugars and other metabolites are also reduced in addition to the decrease motility & food intake in rats. The results have been discussed between the standard and treated rats.

Key words: Colchicine, meadow-saffron,plant alkaloid, gastrointestinal malfunctioning, mucosal cells, nacrosis

INTRODUCTION

Colchicine is a naturally occurring alkaloid with weak anti-inflammatory activity derived from the meadow saffron (*Colchicum autumnale*) and therapeutically used extensively in the treatment of gout for many centuries and also been recommended in preventing attacks of familial Mediterranean fever (Zemer *et al.*)[1] and in the treatment of primary biliary cirrhosis (Kaplan *et al.*,1986)[2]. colchicine is an anti-mitotic agent and has an effective role for acute gouty arthritis. Colchicine interfers with the formtion of microtubule by binding to microtubular and causing depolymerization of microtubules in diverse functions, including cell movement, vesicle transport and chromosome segregation during mitosis. it has been shown that colchicine cab be either cytotoxic or protective against cytotoxicity. Colchicine has been reported to protect against a variety of hepatotoxic insults.Studies on the effecty of malabsorption on the functions of the small intestine would be greatly facilitated if such states could be routinely induced in the laboratory. A number of efforts to achieve such states in experimental animals have met with little success (Althausen and Grodsky, 1961;Williams and Laster, 1963)[3,4] but the use of antimitotic drugs, such as aminopterin and colchicine, cause a pathological villous architecture similar to that found in many types of malabsorption (Williams and Laster 1963; Baserga and Morsiani, 1958; Clark and Harland, 1963)[4-6]. Studies on the effect of colchicine on intestinal absorption have, however, only been carried out in conscious animals (Clark and Harland 1963, Luketic and Shapiro, 1964)[6,7] where absorption depends not only on the activity of the mucosal cells of the intestinal epithelium but also on factors such as gastric emptying, intestinal motility, blood flow, etc. In

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order to avoid the influence of a number of these factors the effect of colchicine on intestinal function has been investigated using a recently developed technique in the anaesthetized rat in vivo and on sacs of isolated, everted intestine incubated in vitro.

MATERIALS & METHODS

White, female rats (200-220 g.) of the Sheffield strain were injected intraperitoneally with colchicine (British Drug Houses, laboratory reagent) made up freshly each day by dissolving 10 mg. in 10 ml. of 0-9% sodium chloride; 0-1 mg./100g. body weight was injected intoeach rat each day for three days. Control rats were injected with similar volumes of 0 9 % sodium chloride. During treatment both control and colchicine groups were allowed access to tap water adlib but no food was given.

Absorption technique

Experimental design

The animals were divided into four groups of 4 each according to the time of sampling: (Group 1): control group, the animals received i.p. injection of saline. (Group 2): The animals were sacrificed at one day post treatment with colchicine. (Group 3): The animals were sacrificed at four days post treatment with colchicine. (Group 4): The animals were sacrificed at seven days post treatment with colchicine.

divided into five equal parts. The middle fifth, usually 16 cm. long, was taken for incubation in 15 ml. of Krebs's bicarbonate saline containing the sugar under investigation. One millilitre of the same solution was placed inside the everted segment. The incubation was carried out at 38°C. for 30 minutes with the buffer in equilibrium with 95% 02, 5 % CO_2. Glucose and other reducing sugars were estimated by the method of Nelson (1944)[8] as modified by Somogyi (1945)[9]. Fat-free dry weights of small intestine were obtained by removing the combined jejunum and ileum and drying overnight at 100°C. The tissue was then extracted with ether, redried, and weighed.

Histological analysis

Rats of each group were sacrificed at 1, 4 and 7 days after treatment with colchicine. Animals were sacrificed by decapitation, the intestine, were excised and fixed in 10% formalin, dehydrated in ascending grades of ethyl alcohol, cleared in xylol, embedded in molten paraplast at 56°C and cut at 5µ on rotator microtome. The paraffin sections were stained with haematoxylin and eosin (Drury and Wallington, 1980)[10]. Histopathological studies were undertaken through light microscopy and photomicrographs were made.

RESULT

Liver Examination

Intestine of control, untreated rat revealed no histopathological changes. No changes were detected in examined sections of rat treated with colchicine at one day post treatment. At four days post treatment with colchicine the intestine revealed activation of mucous secreting epithelium and glands with hemorrhagic area. Hyperplasia and hyperactivation of mucous secreting cells, mononuclear cells infiltration and edema in lamina propria with large degenerative areas containing cellular debris and small hemorrhagic areas were observed in examined intestine treated with colchicine seven days post-treatment.

DISCUSSION

Colchicine is a alkaloid when taken in heavy doses caused serious threating to life. But when taken safely according to the guidelines provided does not proves to be injurious to health but causes serious systemic effects if ingested in doses that exceed the recommendations. In the present study, effect of colchicine was observed in the different tissues of the intestine of the albino rat. The administration of colchicine resulted in histopathological changes in the liver of rat manifested by sporadic necrosis, pyknosis and vacuolations of some hepatocytes, leucocytic cells infiltration, and edema of portal tract with new bile ductules, corrugated portal vein surrounded by fibrotic area, hepatocytes with foamy appearance, dilatation and congestion of hepatic sinusoids, hepatic haemorrhage, loss of hepatic architecture and mononuclear cells infiltration replacing focal areas of hepatic necrosis. They summarized that colchicine had only weak antioxidant properties, but afforded some protection against oxidative stress; more importantly, long treatment with this drug may be of value in producing regression of established cirrhosis. It is reported that colchicine inhibits taurodeoxychlate transport in pericentral but not in periportal hepatocytes. Also colchicine administration revealed hyperplasia and hyperactivation of mucous secreting epithelium and glands and mononuclear cells

infiltration and edema in lamina propria with large degenerative areas containing cellular debris and small multihemorrhagic areas in examined intestine of rat. Hampton (1966)[11] stated that the intestinal pathology after large doses of colchicine is probably related to cessation of cellular proliferation and has been compared to that observed in acute radiation injury of the bowl. Hawkins *et al.* (1965)[12] and Finger and Headington (1963)[13] reported that heavy doses of colchicines resulted in inhibition of growth of cellular elements of crypts. Thus colchicine when taken in heavy may also lead to malabsorption as well as complete lose intestinal villi with concomitant steatorrhea and xylose malabsorption, as well as a depression in activity of disaccharides in the intestine of man . Previous studies were made on colchicine toxicity and poisoning due to over doses or accidental ingestion. Yamada *et al.* (1998)[14] made a histopathological study of experimental acute poisoning of cattle by autumn crocus (*Colchicum autumnale* L.). They reported that all the calves developed severe diarrhea and died or euthanized within 63 hr. and at necropsy the gastrointestinal mucosa was edematous and hemorrhagic and histologically, necrosis and degeneration with karyopyknosis and karyorrhexis were shown in the basal cell layer of the tongue, esoghagus, forestomach, renal pelvis, urinary bladder, neck cell layer of the abomasal gastric glands and intestinal crypts. They also reported that the lesions of the present acute crocus poisoning of cattle closely resembled those reported in humans with colchicine intoxication. It is also observed when colchicines is taken for the extended time period may also lead to malfunctioning of the gastrointestinal tracts (Maxwell *et al.*, 2001)[15]. Currently there is no specific treatment is commercially available for the treatment of colchicine toxicity. Colchicine specific Fab fragments consist of the light chain and variable region of the heavy chain and are derived from goats (Kaplan *et al.*,1986)[16].

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Effect of an organophosphate pesticide in the ovulation of air breathing fish *Anabas testudineus* (Bloch, 1792)

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Abstract : The pollution of environment due to use of pesticides has become an increasing problem over the last century with the development of industry, agriculture and increase in population. The organophosphate pesticides are widely used because of their rapid biodegradability and non-persistent nature. Changes occurring in the haematological characteristics provide a sensitive measure to assess the fish health. Pesticide is a major factor for fish mortality along with water temperatures. The main purpose of the study was to show the vital effect of organophosphate pesticides in the ovulation of air breathing fish.

Keywords:- Organophosphate pesticides, *Anabas testudineus,* biodegradability.

INTRODUCTION

Pesticides as a group of environmental contaminants cause severe toxicity, both acute and chronic in non-target species. Pesticides are a group of environmental contaminants causing severe toxicity, both acute and chronic in non-target species. The word "organophosphates", when appearing in communications (e.g., from the press or the government), in areas such as agriculture, the environment, and human and animal health, very often refers to a group of insecticides (pesticides) that act on the enzyme acetylcholinesterase. Today, organophosphates make up about 50% of the killing agents in chemical pesticides.[1] Organophosphate pesticides (OPPs), like some nerve agents, inhibit this neuromuscular enzyme, which is broadly essential for normal function in insects, but also in humans and many other animals. Literature abounds with reports on the impact of pesticides on various fish tissues but studies on reproductive toxicity in fish are not as many. Studies on pesticidal effects on female reproduction are abortion in *Gambusia affinis* (Boyd 1964)[1], reduction in reproductive efficiency in brown trout, *Salmo trutta,* and brook charr, *Salvelinus fontinalis* (Burdick *et al.* 1972)[2], prevention of reproduction by Carbaryl (Carlson 1972)[3], reduced hatching of embryos by PCB in brook charr (Freeman & Idler 1975)[4], retardation of ovarian development by Fenitrothion and Carbaryl (Saxena & Garg 1978)[5] and Phenthoate (Dey & Bhattacharya 1989)[6] in *Channa punctatus* and by Malathion and Endrin in *Heteropneustes fossilis* (Singh & Singh 1980)[7]. Recently we reported the reproductive toxicity of Metacid-50 pituitary GtH content and hypothalamic GnRH activity in *C. punctatus* (Ghosh *et al.* 1990)[8]. The present work is a follow up of our earlier observation to evaluate

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the effects of non lethal concentrations of metacid- 50 (Methyl-parathion 50%) and carbaryl on the ovarian weights (GSI) and levels of estrogen in the ovary and plasma of the Indian climbing perch, *Anabas testudineus* (Bloch) during an exposure regimen of 90 days extending over the pre-spawning and spawning phases of the annual reproductive cycle.

MATERIALS & METHODS

Pesticides

Metacid-50 is a commercial organophosphate containing 50% methyl-parathion as the active ingredient. Carbaryl is also a commercial formulation containing 50% n-methyl naphthyl-1-carbamate as the active ingredient.

Healthy fishes were procured from local fish market and were kept in cemented tank. The water depth of 0.5-0.75 m was maintained in the tank. Fish were fed twice daily, i.e., 11 am and 6 pm. Perfectly mature female fish have swollen, reddish vent and ooze eggs while the males ooze milky milt upon slight pressure in the abdomen.

Treatment

The tolerance limit for Metacid 50 was 5.33 ppb and that for Carbaryl was 15.83ppm. Depending on the mortality rate & phusiological distress the exposure doses was selected. Since the experiment was carried out for longer duration the fishes were once daily. As per Anon, (1960)[9] regulations the water was changed daily and 0.106ppb of Metacid-50 and 1.66 ppm of Carbaryl were at first dissolved in 20 ml of the aquarium water and then added freshly to negate the effect of metabolites on the toxicity of the poison. the experimental was done for the determination of GSI and hormonal profiles on day 3,7,15,20.

The gonads were dissected out and weighed on a single pan balance. The GSI was calculated on 100g body weight on the basis of given by Abidin (1986)[10].

Collectiojn of blood was done from caudal region and by the process of centrifugation plasma was separated at 2500 g at 4° C for 10 min. The ovary was disected very carefully and immediately homogenized in 0.6% saline; after centrifugation the supernatant was decanted and stored at 4°C for future use.

Radioimmunoassay of l7P-Estradiol

Extraction efficiency was more than 85%. Anti-Estradiol-17/3 was diluted 1:25000 in phosphate buffered saline-gelatin (PBSG). 100p1 of antiserum were added to the standards which ranged from 10 to 1000pg. After the incubation of the samples for 1 h, of 3H-E2 were added to each tube. Samples were incubated at 4°C for a night and then placed in an ice bath for 15min. Samples were then centrifuged, decanted into scintillation fluid. The results were expressed as mg per ml plasma and pg per mg of ovarian protein. Samples were then prepared and stored at -24°C til the end of the sampling period.

RESULT AND DISCUSSION

When the fish *Anabas testudineus* was treated with the carbaryl 15 days after exposure there was no deviation from the control value of GSI (Gonadosomatic index). But however when it was treated from 20 days onwards until the termination of the experiment there was great variation in GSI level from the respective control value. The profiles of the plasma and ovarian estradiol-17P (E,) of the control fish showed a gradual increase until 60 days of observation. When the ovarian E_2 level was observed 90 days after exposure showed decrease in the level as compared to those experimented for 60b days. In the case of plasma1 & ovarian E, levels in the organophosphate treated fish a gradual increase was observed from day 3 to 15 days of treatment, thereafter showing a sharp decline until day 60 of treatment. However, at 90 days of sampling both plasma and ovarian E level, it showed little improved 87% decline of plasma E_2 and 29% in the ovarian E, level over the control. After 15 days of carbaryl treatment the plasma E_2 titre showed significant increase (18%-128%) while in the later period from 20-60 days of exposure the hormone level decreased (19%-83%) significantly. However, at 90 days, the degree of depletion was not very remarkable. Ovarian E_2 level, on the other hand, showed a small decline on day 3rd day leading to 25-45% increment on day 7 and 15 day, respectively. This increase subsided to only an 8% increase on day 20 which was followed by significant depletion from day 30 to 90.

The exposure of the two different classes of pesticides caused no mortality of the fish or sign of physiological distress. However, they were both potent enough to cause significant reproductive impairment to ovarian tissue. Histological examination revealed a

preponderance of stage I and destruction of stage 11 and stage 111 oocytes in the fish treated with the pesticides (unpublished observation). As reported by Shukla (1982)[11], GSI is greatly affected by DDT, BHC, endosulfan, chlordane and toxaphane. Dey & Bhattacharya (1989) observed the preponderance of stage I and destruction of stage 11 and stage 111 oocytes in association with decreased ovarian weight in phenthoate exposed *C. punctatus*. Previous workers have suggested an impairment of steroidogenic activity in the gonad of pesticide treated fish (Freeman & Idler 1975, Saxena *et al*. 1986, Kapur *et al*. 1978)[4,12,13]. GtH in *Channa punctatus* exposed to sublethal doses of metacid-50 and carbaryl for 30 days. It may, therefore, be suggested that the decreased steroidogenic activity in *Anabas testudineus* during this period may be due to the decreased release of GtH during a long term exposure.

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Morphology of worm snake *Typhlina bramina*

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Abstract : Snakes are one of the most wide spread vertebrates on the globe, mostly they prefer arid zone of the globe in terms of diversity. Data collected based on reports from volunteers, snake friends and rescuers 5 species of snakes *Typhlina braminus* reported from Madhepura region of Bihar. Road kills, encounter during work in agriculture fields and poor knowledge of the people in general about failure in distinguish between the worm snake and earthworms were main reasons behind destruction of snakes from this region.

Keywords:- Organophosphate pesticides, *Anabas testudineus,* biodegradability.

INTRODUCTION

There are several species of snakes which looks likes worms and hence known as worm snake. These worms snake usually shares the characteristics of small size, primarily subterranean habitat, and small eyes and thus resembles like earthworm and therefore sometimes people get confused. These snakes are sometimes also known as blind snake which is also alternative common name. Blind snakes generally belong to the families Anomalepidae, Leptotyphlopidae, and Typhlopidae in super family Typhlopoidea. Since these three families are the only ones classified within infraorder Scolecophidia, blind snakes are sometimes called "scolecophidian snakes." Genera of snakes often called "worm snakes": Blind snakes are mainly small with blunt heads, cylindrical bodies, and short tails. Their bodies are encased in smooth, shiny scales. All are fossorial and are regularly associated with termite and ant nests.

Typhlina braminus (Daudin, 1803)[1], also known as the Brahminy blind snake or flowerpot snake, is a small worm-like snake often sometimes mistaken for an earthworm. It is always among the smallest snakes wherever it occurs. Its mean total length is between 100-130 mm with a weight of 0.75 gm (Wallach 2009)[2]. Based on the presence of retrocloacal sacs (indicated as "retro-cloacal formations" as well as unique features of the hemipenes, Robb (1966)[3] showed that the genus *Ramphotyphlops* Fitzinger, (1843)[4] had to be revalidated to accommodate certain Australasian species. Later on McDowell (1974)[5] noted that *braminus* could be assigned to the genus *Typhlina* Wagler, (1830)[6] (a synonym of Ramphotyphlops) based on similarities with the head scale pattern of "*Typhlina erycina*." A complete nomenclatural history of the species (as *R. braminus*) can be found in Wallach (2009)[2], who indicated that a new genus has to

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be proposed for that species. *Typhlina braminus* is the most successful species in the entire family, spreading on all continents most often in recent years but also sometimes earlier through human involuntary introduction through trading and movement of people and goods around the Indian Ocean and beyond. The snake is, usually not present in South America, and arid, high latitude and high altitude areas. These types of can easily be seen abundantly and easily under objects deposited beneath houses, particularly in villages. The species is terrestrial, but some specimens have been observed in arboreal habitats (Bazzano 2007)[7]. Ota *et al.* (1991)[8] estimated clutch size of this snake as 1-8. Incubation period is of up to 58 days. Parthenogenesis, first suspected (McDowell 1974, Nussbaum 1980)[5,9], was demonstrated by Ota *et al.* (1991)[8]. Its populations only comprise triploid females that reproduce through obligate parthenogenesis, a characteristic unique among snakes (Booth & Schuett 2016)[10]. Some or all parental species could now be extinct through competition with their most successful hybrid descendants. *Typhlina braminus* feeds primarily on the soft parts of ants and termites, sometimes on earth worms and caterpillars (Wallach 2009, Geniez 2015)[2,11]. Recently Mizuno showed that it decapitates its termite prey and consumes only the thorax and abdomen in about half the specimens it eats. Decapitation in worm snakes allows the snake to avoid consuming termite heads which are of poor nutritive value or to escape chemical defense of termites by non-ingestion of toxic compounds mostly located in their heads. These types of snakes have been discovered from various parts of the world. Sri Lanka, China and other Southern parts of the country.

MATERIAL & METHOD

To determine morphology of wormsnake *Typhlina bramina* in various habitats of Madhepura region first the different local areas of this region was selected and hence the identification as well as morphological studies was made during the 2017 to 2018. The study was conducted with the help of local communities residing on selected zone of Madhepura by interviewing and showing different colors photographs of snakes. On the basis of identification by the local communities, check list was prepared as a primary data. The identified Specimens of *Typhlina bramina* were euthanized and preserved in 75 % ethanol in the zoological laboratory of B.N Mandal University Madhepura.

RESULTS & DISCUSSION

During the observation period 5 different species of worm snake was founded which was then further taken to lab for its morphological studies.

Species	Region	Total length((mm)	Body weight
Typhlina braminus	Puraini	60 – 197	0.30 – 1.40
Typhlina braminus	Bihariganj	57 – 200	0.30 – 1.87
Typhlina braminus	Gamhariya	63 – 203	0.30 – 1.00
Typhlina braminus	Kumarkhand	66 – 178	0.30 – 1.40
Typhlina braminus	Ghelardh	43 – 161	0.80

Species	Region	Total middorsals	Caudal scales	Total vertebrae
Typhlina braminus	Purainai	261 – 341	11 – 15	175 – 192
Typhlina braminus	Bihariganj	261 – 344	12	180 - 200
Typhlina braminus	Gamhariya	281 – 368	11 – 13	187 – 206
Typhlina braminus	Kumarkhand	296 – 336	8 – 13	182 – 195
Typhlina braminus	Ghelardh	306 – 360	10	179 – 196

Morphology

Typlina braminus is regarded as of the smallest snake among the different worm snakes found in the world and also smallest among the snake fauna in appearance it looks same like that of the earthworm and therefore it is sometimes mistaken as an earthworm**.** Blind snake are characterized by degenerate eyes that lie beneath opaque head scales.

Size and proportion

There are various claims made regarding size of the *Typhlina braminus* in the literature. Unconfirmed reports listing the maximum length as 220–230 mm which may be due to misidentification by Smedley of a 230 mm *Ramphotyphlops albiceps* that was reported as a *Typhlops braminus*. Numerous reports of lengths between 170–190 mm indicate that *T. braminus* occasionally reaches such lengths.

These specimens have not been examined to confirm identification but they are provisionally accepted and indeed represent giants if correct. According to the above observation it was found that the species collected from the Gamhariya region was of the maximum length about 63-203 having maximum weight about 0.30-1.40gm while the species that was smallest in length was observed from the Ghelardh region which was about 43-161 mm in length having weight of about 0.80 gm.

Colouration

The colour pattern of the *Typhlina braminus* usually varied from dark brown to black as well as white in colour. Ventral and dorsal region varied from black to brown while the tip of the tail region was found to be white in its colourations.

Nostril

Snout is usually rounded laterally and dorsally, rostral usually oval and narrow (0.25–0.33 intraocular head width), with anterior constriction visible dorsally; supraocular oblique, twice as long as broad; frontal, postfrontal and interparietal smaller than supraocular, occasionally transversely enlarged; parietals transverse, enlarged and occipitals occasionally enlarged.

Eyes

Eye is usually small, with distinct pupil, partially or nearly located under the lateral border of the supraocular. A postocular is single; twice the height of the costal scales, fourth supralabial is usually larger than other three supralabials, with a posterior notch suggesting their fusion of another scale with the supralabial.

Head

An anterior head shield usually contains distinct rows of whitish glands beneath their posterior borders, with the prenasal gland line confluent with the posterior rostral gland line dorsally.

The glands have been described as forming "a faint crenellated whitish marginal line". Additionally, there are numerous integumantarey sense organs that cover the shields of the head but most heavily concentrated on the anterior surfaces of the head.

Distribution of shields is as follows: superior nasal, inferior nasal, rostral, supralabials I-III, supralabial IV, infralabials, supraocular, frontal, postfrontal, parietal, interparietal, ocular, postocular. Costal scales are often imbricate and dorsally cycloid; ventrally the scales usually appear trapezium in shape with nearly point contact. Tail slightly longer, apical spine with stout base is directed horizontally; near base of bifurcation tongue is present with a pair of lateral papillae.

Vertebrae

The total number of vertebrae observed during investigation varied depending upon the species collected from the particular region.

The highest number of vertebrae ranged from 187-206 species of *Typhlina braminus* collected from Gamhariya region while the species collected from the Purainai region had the total number of vertebrae (including axis, atlas, trunk, cloacal and caudal) ranging from 175-192.

CONCLUSION

On the basis of various observation, findings and identification it can be concluded that the *T. braminus* is one of the species belonging to worm snakes variety which usually resembles like those of earthworm and therefore sometimes it is also mistaken as an earthworm. *Typhlina braminus* is also by its name *Indotyphlops braminus* or *brahminy* blind snake. It is one of the smallest species of among the different species of snake variety.

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Effect of temperature on the population density of aquatic insects

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Abstract : The population growth of *Cybister confusus*, an aquatic insect in a pond was studied with emphasis on water temperature. Population of the *Cybister confusus* was examined in relation to degree-days. This measure accounts for time and the minimum temperature necessary for insect growth and reproduction to enhance population size. Result showed highest population density at about 17.6°C temperature. Studies demonstrated that the population of insects is directly proportional to change in seasonal temperature cycle

Keywords:- ***Cybister confusus,*** **population, Madhepura, temperature**

INTRODUCTION

The most diverse and prominent order of the insect, Coleoptera includes about 3.87 lakhs described species from the entire world representing about 38% of insect species (10.2 lakhs species) of all the orders[1]. Out of these, 12,604 species belong to aquatic beetles pertaining to at least eighteen families[2]. Even though the insects are terrestrial in origin, a large number of species belonging to several orders have adapted to aquatic mode of life. Despite a lot of fresh water bodies in India, the information on these aquatic beetles in several states is still lacking. The catalogues and review work on aquatic beetles of world are published by Schenkling (1910-40)[3], Knisch (1924)[4], Wewalka (1975 & 1979)[5,6], Brancucci (1983)[7], Pederzani (1995)[8], Miller (2002)[9], Andrew and Martin (2011)[10], Nilsson (2011)[11] (palcat.) & 2013 (world cat.)[12], Fery (2003)[13], Toledo (2009)[14]. The information on aquatic beetles of India is known through the works of D'Orchymout (1928)[15] and Vazirani (1969a, 1969b, 1977, 1984)[16-19] where in the species from Central India was also included. Subsequently, Chandra *et.al.* (2010)[20] and Ghosh & Nilsson (2012)[21] reported total seventy two aquatic beetle species from Madhya Pradesh.

However, study on the aquatic beetle fauna from Bihar state and particularly in Madhepura districts insufficient. Hence the present work was undertaken. The present study, "Effect of temperature on the population density of aquatic insects" is a part of the work to investigate the impact of environmental stress in the form of temperature on an aquatic insect- *Cybister confusus* in terms of population. *Cybister confusus* belongs to the class-insecta, order-coleoptera, family-dytiscidae, sub-family-dytiscinae and genus-cybister.

It was hypothesized that two factors, temperature and time, would be sufficient for predicting the aquatic

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insect populations. Previous studies have demonstrated the degree-day dependence of insects' growth[22-24]. Few, however, have examined the extent to which the degree-day relationship holds within a given habitat. Harper (1973)[25] notes that differing temperature regimes, within the habitable range of an organism, can yield differences in growth patterns of stoneflies. Studies with *Assellus* spp. show very definite growth-temperature relationships using the degree-day concept[23, 24].

Water beetles can be used to control water plants that have become pests. Species of Dytiscidae are aquatic predators and may play an important role in controlling mosquitoes. Dytiscids cause considerable harm to fish fry but there were few actual studies on that subject, and more research would be needed to assess potential harm as well as benefits of water beetles to aquaculture[26]. The earlier knowledge and scientific contribution on aquatic beetles are noteworthy to understand the present fauna[19,27,28]. Besides thisa number of other workers contributed greatly[2,5]. The major studies on aquatic Coleoptera also includes the works from Andhra Pradesh (Mukhopadhyaya, 2007)[28], West Bengal, (Biswas &Mukhopadhyay, 1995)[29], Sikkim (Mukhopadyaya & Ghosh, 2003)[28-30].

MATERIALS AND METHODS

A pond at Babhangama village was selected in Bihariganj block of Madhepura district (Bihar) to study the aquatic insect- *Cybister confusus* population for a period of 12 months from May, 2017 to April, 2018. Monthly collections were made for the entire period of study in three stations in the pond. Pond has an area of about 30,000 Sq. ft. with an average depth of five ft.

Description of Stations

Station-I: Dense with floating and submerged vegetation.

Station-II: Open water area completely devoid of vegetation and used as a bathing ghat.

Station-III: Open water area with sparse vegetation and the deepest area of the pond.

Collection and calculation Methods

Preliminary sampling was initiated in May, 2017. Regular sampling, at 30 days intervals, was begun after that. According to Menke (1979)[31] the aquatic bugs in particular are to be collected differently according to their behaviours and dragged through the organic debris, floating vegetation, tangled roots and other objects. Insects clinging on the vegetation were handpicked. Some insects were collected by splashing water on the bank toflush them out of the crevices and small insects that hide in mosses and floating vegetation were exposed by pressing the plant under water. The aquatic insects in open area were collected with the help of a pond net.

During the course of study, surface floating/ swimming insects were collected with small circular nets made of finely meshed polyester mosquito curtain cloth. The design and operation of the net was roughly based on those described by Junk (1977)[32]. The insects were collected by filtering forty litres of water randomly from each station with the help of a boat and pond net.

Habitat sampling of the insects and water were made during early hours of the day (7 am-10 am) since many aquatic insects migrate to deeper water during late hours of the day. The adult insects collected were sorted out for *Cybister confusus* after identification with the aid of standard literature on the group, kept proper record of the identified population. After proper record keeping insects were released in same pond. Along with sample collection temperature also measured every time with the help of laboratory thermometer.

RESULTS AND DISCUSSION

Dissolved oxygen, which is inversely correlated to temperature, decreases with increasing temperature and increases with decreasing temperature[33]. In this case, the dissolved oxygen values are expected to be high in the winter months. The relationship was clearly seen in present study. While the lowest temperature value of the pond water was measured as 11.8°C in January 2018 and the highest temperature value was measured as 35.9°C in July 2017 (Table-1).

As per the change in temperature of the pond water population statistics also changed. There was a strong negative relationship between temperature values and population of the species. This data show us that high temperatures are not suitable for aquatic insects particularly, *Cybister confususn* and as the temperature approaches the upper limit, the population of species was decreased. However, the highest population density value was seen at 17.6°C in March 2018.

Table-1: Population of *Cybister confusus* in a pond at Babhangama village, Bihariganj block of Madhepura district (Bihar)

Sl. No.	Month	Station-I		Station-II		Station- III		Mean value	
		Temp. (in °C)	No. of individual (in 40 L)	Temp. (in °C)	No. of individual (in 40 L)	Temp. (in °C)	No. of individual (in 40 L)	Temp. (in °C)	No. of individual (in 40 L)
1	May,17	28.5	134	26.3	44	27.5	195	27.4	243
2	Jun, 17	32.8	122	31.7	32	30.4	172	31.6	109
3	Jul, 17	34.8	94	36.8	22	36.2	122	35.9	157
4	Aug,17	32.2	85	35.8	21	34.2	103	34.1	140
5	Sep, 17	31.9	92	34.0	18	32.2	95	32.7	142
6	Oct, 17	28.9	88	31.5	07	29.4	106	29.9	130
7	Nov,17	22.4	105	25.2	19	23.1	128	23.6	167
8	Dec, 17	18.1	117	20.4	36	17.1	133	18.5	197
9	Jan, 18	11.1	152	13.2	42	11.2	137	11.8	240
10	Feb, 18	15.4	174	15.2	49	15.6	179	15.4	283
11	Mar, 18	17.9	193	17.3	57	17.5	216	17.6	322
12	Apr, 18	23.3	184	21.6	49	22.8	191	22.6	297



Fig. 1- Pond at Babhangama village, Bihariganj block of Madhepura district (Bihar)

CONCLUSION

To study population and temperature relationships, it is necessary to determine whether or not a minimum critical temperature for a species is reached in the water bodies. Below this temperature, population is either very low or non-existent[34]. In present study critical temperature was found about 17.6°C. This temperature minimum is difficult to evaluate with field data, but an approximation is possible. Further studies aiming to improve our knowledge on water insects should focus on collecting in little known areas, revision of the still unstudied material from additional families and filling the large gaps in our

knowledge regarding the diversity of water beetles in some specific habitats.

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A taxonomic study on the aquatic insect of order coleopteran found in Jharkhand

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Abstract- 95 species of aquatic insects belonging to order Coleoptera have been recorded from a weed infested man made wetland water bodies in different parts of Jharkhand. Coleoptera was numerically the most abundant group comprising 39% of the total aquatic insect followed by Hemiptera (36%) and Odonata 25%. And the variation may be due to unfavourable water condition for the survival of the beetles due to pollution.

Keywords:-Aquatic insect, biodiversity, coleoptera, ecosystem

INTRODUCTION

Fresh water ecosystems are colonized by a diverse array of aquatic organisms. Aquatic insects, a group of arthropod, live their life cycle in water bodies and are found in or on the surface of the lentic or lotic water. There are 8 major orders of insects that spend at least a part of their life cycle in the fresh water ecosystem. Aquatic insect are considered as a very good indicator of water quality because of their wide range of environment disturbance tolerant levels. Some of these insects may be beneficial to human beings, while few others may be quite harmful. These insects form an important component of the food chain and energy flow pathway and comprises of a high proportion of biomass in fresh water ecosystem. Aquatic insects are used for assessing water quality and provide information to environmental managers to take accurate and justifiable actions in regards to the state and quality of water bodies. Many groups of aquatic insects are susceptible to pollution or alteration of their habitat. The presence or absence of aquatic insects can indicate whether a particular system is healthy or polluted pond often constitute biodiversity "hot spot" within the area. The information on aquatic beetles of India is known through the works of D'Orchymout (1928) and Vazirani (1969a, 1969b, 1977, 1984), where in the species from Central India were also included. Subsequently, Chandra *et al.* (2010) and Ghosh & Nilsson (2012) reported total seventy two aquatic beetle species from Madhya Pradesh.

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MATERIALAND METHODS

Study area

The field work was carried out on the different parts of selected areas in Jharkhand such as Ranchi, Khunti etc. the pond rivers and the rivers were selected for the purpose of study. In Ranchi Badatalab, Karamtoli tallab, Kanke dam etc was selected which is located in northern and eastern side of the Ranchi University.

Data were collected by four fields between November 2017 and February 2018. In each of the subdivisions, villages were randomly selected and surveyed through the help of participatory rural appraisal (PRA) tools like semi structured questionnaires, interviews, and field observations. In each of the villages sampled, at least 30% of the populations were surveyed. Respondents were basically men and women of age above 18 years. In total, 350 semi structured questionnaires were distributed. In case of grouped interview, open oral discussion on issues related to coleoptera consumption was engaged with the group. In either case, questions were asked on the consumption (larvae and adults).

Field work

Field work was conducted in rural areas of Jharkhand. This included regular outings spanning four seasons, with a stay of at least fifteen days per season in several areas. The aim of the fieldwork was to conduct semi-structured interviews of the ethicist type (is a technique for a person to convert orally interviewer personal definition of the situation). The inhabitants of various localities were interviewed concerning the use of water beetles in their food. Interviews were carried out in the street markets, or on the streets or even in homes. The interviewees were adults of both sexes from 20 to 65 years of age.

Tools used

Tools generally used for the collection of the species were the planktonic net.

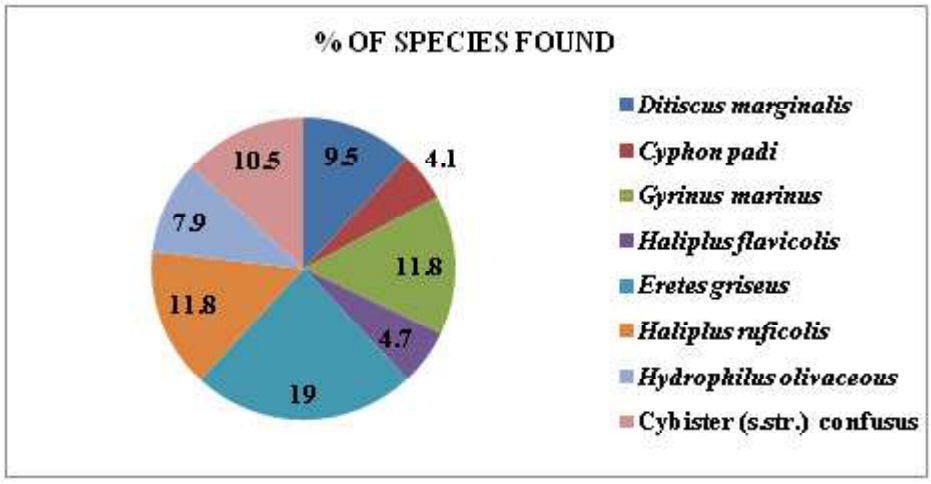
Sampling method

Random sampling of the local insect related to the study area was done during the period of 2017-2018 representation of each group and order was spread

Table 1- Shannon Weiner diversity index

SL. no.	Name of the insect group	Individual of species	Abundance pi=n/N	Log pi	pi*logpi H=pi+log pi
1	*Ditiscus marginalis*	10	0.10	-1	-0.1
2	*Cyphon padi*	23	0.24	-0.61	-0.14
3	*Gyrinus marinus*	8	0.08	-1.09	-0.08
4	*Haliplus flavicolis*	20	0.21	-0.67	-0.14
5	*Eretes griseus*	5	0.05	-1.3	-0.06
6	*Haliplus ruficolis*	8	0.08	-1.09	-0.08
7	*Hydrophilus olivaceous*	12	0.12	-0.9	-0.10
8	*Cybister (s.str.) confusus*	9	0.09	-1.04	-0.09

Graph 1- Percentage of species found during 2017-2018



preserved after proper pinning in the insect collecting box of the department laboratory. Appropriate photography of the sampling site as well as insect residing in that habitat was also done. Detailed taxonomic study was done in the laboratory with the help of Imm's taxonomic of each other. During the field sampling the numerical counting of the other individual of the same species in the particular quadrant was also done.

RESULTS AND DISCUSSION

From the 6 months of observation it revealed that the percentage of occurrence of *Haliplus flavicolis, Haliplus ruficolis* belonging from the family Halipidae with respect to other aquatic coleopteran families were the highest among the demonstrated areas of Ranchi district. Among the remaining species belonging to family Dytiscidae, Scirtidae, Gyrinidae and Hydrophilidae were found in all the water bodies but few in numbers.

CONCLUSION

Five families namely Dytiscidae, Scirtidae, Gyrinidae, Halipidae and Hydrophilidae are chiefly represented in the present report. The study opened the opportunities for future taxonomic and diversity studies on the aquatic Coleoptera of Jharkhand state.

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Species diversity of shellfishes in Madhepura district of Bihar

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Abstract- The abundance and distribution of some shellfishes of the coastal waters bordering different areas of Madhepura district of Bihar was studied between September and December, 2018. Specimens were collected monthly with the assistance of local fishermen. The population structure and distribution patterns were determined using Shannon-Weiner diversity index. The species population diversity showed great fluctuations in all the four different sampling sites during the different months of the observation period. All most of the shellfish species were found to be abundant during the months of September and October while during the other study period the species were present in few of the sampling site which may be due to high pollution status.

Keywords:- Shellfishes, Shannon-Weiner diversity index, Madhepura.

INTRODUCTION

Madhepura district is one of the thirty-eight districts of Bihar, India. Madhepura town is the administrative headquarters of this district. Madhepura district is a part of Kosi division which has rich biodiversity that contains a diverse assemblage of fish, shellfish (shrimps, crabs, lobster, gastropods and cephalopoda), reptiles and other living organisms. Shellfish harvested by the artisanal fishermen include white shrimps (*Nematopalaemon hastatus*), brackish river prawn (*Macrobrachium macrobrachion*), *Bellamya bengalensis*, *Pila globosa* etc. Shellfishes have been found out to be of very great commercial importance in most of the country. And thus the exploitation has been on the increase due to increasing population and increase in demand for protein sources by man[1]. The flesh of *Pila* and *Viviparous* sp. is used as medicine to cure asthma, swelling of joints, burns by aboriginal people of this region. The foot portion of these edible shell fishes is eaten in the form of curry as well as in roasted form at regular intervals. The shell fishes and crustaceans are farmed commercially and dominate world output of aquacultural resources. Among shell fishes, the pearl Oyster, *Unio*, *Pila*, dominates in one hand and crustacean's crabs (*Paratelphusa spinigera*) and prawns (*Macrobrachium* sp.) on other hand. Fish farming is a solution to the world food problem source of low cost animal proteins for lower income populations. Nwosu reported that there has been a significant reduction of the natural stock of shrimps in Nigerian coastal waters (probably due to environmental degradation which is detrimental to the abundance and life cycle of the shrimp species) while Deekae and Abowei stated that the unfriendly fishing methods of local fishers who use poisons and chemicals are affecting the shell fish population abruptly. [2, 3]

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MATERIALS AND METHODS

Description of Study Area

The study was carried out in Local area of Madhepura between September, 2018 and December 2018. The shrimps were collected on monthly basis between the months of September and December 2018, with the assistance of artisanal fishermen. The Shrimpers used boats with sizes ranging from 5 to 12m long. The boats were powered by small outboard engines and manned by an average of two men per boat, to which the shrimping nets are attached. The specimens were immediately preserved in iced packed cooler and transferred to the Fisheries and Aquaculture Laboratory, B.N Mandal University of, Madhepura where they were frozen at -4^{0}C before they were used for the research work. The specimens were sorted into different groups and identified to specific level using the FAO Species Identification Sheets.

Data Analysis

The data was analyzed using the Shannon Weiner diversity index by using the formula:

$$H = \Sigma\ pi \log pi$$

Where H = species diversity

pi = mean of individual species procured by the formula n/N

log pi = is the log product of po(mean of individual species

OBSERVATION

Table 1: Occurrence of Shellfishes in coastal waters areas of Madhepura district, between September and December, 2018.

Months	Species	Haraili	Baiwah	Budhma	Khagaria	% occurrence
September	*Sartoriana spinigera*	+	+	+	+	100
	Bellamya (Viviparous) variata	+	_	_	_	25
	Bellamya bengalensis	+	_	+	_	50
	Indoplanorbis exustus(Deshayes)	_	+	+	_	50
	Pila globosa	+	+	+	+	100
October	*Sartoriana spinigera*	+	+	+	+	100
	Bellamya (Viviparous) variata	+	_	+	_	50
	Bellamya bengalensis	+	+	+	_	75
	Indoplanorbis exustus(Deshayes)	+	_	_	_	25
	Pila globosa	+	_	+	_	50
November	*Sartoriana spinigera*	_	_	_	+	25
	Bellamya (Viviparous) variata	_	+	+	_	50
	Bellamya bengalensis	+	_	_	+	50
	Indoplanorbis exustus(Deshayes)	_	+	+	_	50
	Pila globosa	+	_	_	_	25
December	*Sartoriana spinigera*	_	+	_	_	25
	Bellamya (Viviparous) variata	+	_	_	+	50
	Bellamya bengalensis	_	_	+	+	50
	Indoplanorbis exustus(Deshayes)	_	_	_	_	0
	Pila globosa	+	_	+	_	50

RESULT & DISCUSSION

The present study on the species diversity of shell fisheries of the area of North-Bihar reveals that the most of the areas of this region was rich in diversity of molluscan and crustacean fauna. The abundance of shell fishes in terms of species diversity indicates a good life support system for fishes and birds. Shell fishes are the major component of the macro-invertebrates, they form link between zooplankton and vertebrate taxa, such as fishes and birds and play a key role in the energy flow and bio-geochemical cycle of the wetland habitats. A number of fish and avian fauna diversity directly depend upon the mollusks population of these habitats.

The occurrence of the shellfishes in the sampled stations is shown in Table 1. The table shows that *Sartoriana spinigera*, and *Pila globosa* were caught in all locations throughout the sampling period during the month of September while *Indoplanorbis exustus* (Deshayes) was caught only from two sampling site during the month of and *Bellamya (Viviparous) variata* was only observed in Haraili. *Indoplanorbis exustus* (Deshayes) were not caught from the Haraili. However during the month of October the species of shell fishes caught were lower than those obtained in the month of September.

Sartoriana spinigera was collected from all the sampling sites that were chosen for the study of species diversity. *Indoplanorbis exustus* (Deshayes) was present only in one of the sampling site that resembled the lower in species diversity. During the month of November *Bellamya (Viviparous) variata*, *Bellamya bengalensis* and *Indoplanorbis exustus* (Deshayes) were found in only two of the sampling sites.

While duering the month December most of the shell fishes species were found to be in less in number. The *Indoplanorbis exustus* (Deshayes) species were almost absent in all the four sampling site. While the other species were present in very low in number and were found only in few sampling sites of Madhepura district.

The result reveals that species diversity is a useful parameter for the comparison of communities under the influence of biotic disturbances or to know the state of succession and stability in the community as the diversity index. This indicates little difference in the species diversity of the shellfish communities. Balloch *et al*[5] found the diversity index to be a suitable indicator of water quality, Hughes[6] also concluded that this index was useful for community structure, but could not stand alone for assessing environmental quality while Costa and Fransozo[7] reported that the important factors that affect distribution and abundance of shrimps in the tropical region are water quality (temperature, salinity), nourishment and substrates.

Thus, this finding could be as a result of the prevailing highly similar environmental conditions as reported by Ekta[8]. Some important factors governing the abundance and distribution of aquatic communities includes, water quality, immediate substrates for occupation and food availability[9] and any ecological imbalance arising from any severe alterations of these factors may affect the environment. Also, Kennish[10] noted that anthropogenic activities could lead to the periodic or permanent elimination of estuarine dependent fish species from individual estuarine systems. Therefore, the relatively low composition and diversity may be as a result of stress imposed by land based pollutants, as well as substrate instability possibly arising from frequent anthropogenic activities in the area. Similar observations were made by Ajao and Fagade[11] on the western industrialized parts of Lagos Lagoon, which received a complex mixture of domestic and industrial wastes.

CONCLUSION

Above studies done on species diversity of the shell fishes concludes that the *Sartoriana spinigera and Pila globosa* were found to be very abundant and widely distributed throughout the various coastal areas of the Madhepura district. Although there were generally very high similarity among the shellfish communities in the four sampling station. However in few sampling stations the populations of the shellfishes were low due to pollution status. Thus there is very serious need to control and monitor to attain the fishes sustainable diversity.

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Effect of organophosphate on testicular cycle of male *Anabas testudineus*

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Received : 17th October, 2018 ; Revised :19th January, 2019

Abstract- A one year study was conducted from January-December 2018 to investigate the reproductive biology of Indian Climbing Perch in earthen pond. The female fish elucidates rounded, swollen abdomen and reddish pelvic fin while the male fish also possess a reddish pelvis fins but generally smaller, narrower. The surfaces of male's pectoral were little rough during the breeding season. The Gonado-Somatic Index (GSI) value of female *Anabas testudineus* was ranged from 12.71±0.73 (July) to 1.13±0.10 (October) while Hepato-Somatic Index (HSI) value of female was ranged from 4.37±0.64 (January) to 1.58±0.19 (July). The GSI and HSI were inversely related with each other. The fecundity was counted from the month of April to July by using gravimetric method where absolute fecundity during April was 16832.80± 673.34 and during July was 46186.14±2219.15. The relationship between gonad weight and fecundity, body weight and fecundity showed strong positive relation with regression co-efficient of 0.9603 and 0.9265 respectively which indicate that with the increase of body and gonad weight, fecundity was also increased. Microscopic observation of matured female ovary during June to July revealed group asynchronous development. Microscopic observation on matured testes found spermatocytes, spermatids and spermatozoa stages during the month of June and July.

Keywords:- Ovary, fecundity, GSI, HSI, *Anabas testudineus*.

INTRODUCTION

Climbing perch *Anabas testudineus*, locally known as Koi, is an Indigenous air breathing freshwater species. This fish is suitable for cultivation and is highly recommended for its supreme nourishing quality and prolong freshness even out of water (Potongkam, 1972)[1]. It is a predator, carnivore (Pandey *et al*., 1992)[2] or an insectivore fish (Ahyaudin, 1992)[3]. Flesh of *A. testudineus* is rich in iron and copper that support hemoglobin synthesis (Saha, 1971; Sarma *et al.,* 2010)[4,5] and it has high quality poly-unsaturated fats and many essential amino acids. In recent times, because of higher value and market demand this species prone to getting nearly extinction. This might also be due to environmental changes and over fishing (Sverdrup, 2002; Das *et al.,* 2009)[6,7]. Knowledge on reproductive events is necessary to obtain information about the size and age of sexual maturity, spawning season and testicular development. By taking all above into consideration the present study was set out to address some biological aspects of Indigenous Koi. Information on the different biological parameters like body morphometrics and biometrics, HSI, GSI, fecundity are important parameters representing details reproductive

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status and breeding peaks of fish, which provide key information for aquaculture and fisheries management and explain human intervention in aquatic system.

MATERIAL & METHOD

Sample collection and processing

For the study of testicular cycle of *Anabas testudineus* one year study was conducted in the laboratory of Fish Biology. Live fishes were collected from the local market which was more or less same in size. Fishes were then reared in the pond for research purpose up to December, 2018. For experiment, six fish species of male were collected monthly and brought to the laboratory for the observation to measure the total length, body weight, gonad weight, gonadal length of the individual fish. For gonadal observation fish were sacrificed and dissected carefully. Gonads were isolated and weighed by using a sensitive portable electronic balance (XS Analytical105). Then the gonads were divided into three sub samples as anterior, middle and posterior (range between 1g to 0.95g) and then tagged followed by keeping the marked samples in the Bouin's fixative for 20 minutes before the histological process.

Histopathological studies on gonads.

For histopathological investigations, gonads, samples from *A. testudineus* are collected at the end of exposure period. The gonadal samples were preserved in small plastic vials with Bouin's fluid. The samples were then dehydrated, cleaned and infiltrated in an automatic tissue processor. Alcoholic series of higher concentrations, xylene and paraffin wax were used for gonads. The sample were then embedded in paraffin wax and sectioned at a thickness of 5 ìm by a microtome. Sections were placed on a water bath at a temperature of 41°C. Suitable sections were selected from the ribbons which were finally picked up over glass slides. Then the prepared slides were placed on a hot plate over night at a temperature of 42°C for the fixation of the sections. Hematoxylene and eosin were used to stain the sections according to the method described by Toman *et al*. (2012)[8]. The stained sections were then mounted and later examined under a compound microscope. Five histological slides were randomly chosen for each sample. Photomicrographs of the stained sections were taken by using a photomicroscope.

RESULT & DISCUSSION

The histopathological changes observed in the tissues *of A. testudineus* in the present study indicate that sub lethal concentration as well as higher concentration caused moderate to severe alterations in the gonads, which are the important organs performing vital functions. In the histological observation of gonads, structure and systematic arrangement of cells of testis was found normal. At an exposure to 0.001 ppm diazinon, mild fragmentation of testis cell with ruptured wall and karyolysis were found and at the higher doses i.e., 0.032 and 0.064 ppm of diazinon, severe fragmentation of testis cell with ruptured wall and karyolysis were observed.

The highest and most rapid mortality was observed in fish exposed to the highest concentration of diazinon tested. The *A.testuineus* for 96 hrs was found to be as low as 6.55 ppm. Hence, diazinon is considered a severely toxic substance to *A.testuineus as* this fish is very sensitive to the presence of any kind of toxic chemicals in its body. Rahman *et al*. (2002)[9] reported LC50 values of diazinon for breathing fishes, *Chana punctatus* were 6.55 and 3.09 ppm for 96 hrs of exposure, respectively. These values are very high compared to the value of diazinon obtained in the present study for *G. giuris* because air breathing fish have a high tolerance level for diazinon. On the other hand, in case of large sized fishes like *mrigala* and *Cyprinus carpio* the LC diazinon for 96 hrs were 5.8 ppm, 8.15 ppm, and 9.76 ppm, respectively (Rauf and Arain, 2013)[10]. From these studies a similarity was observed in the LC50 values obtained for different larger sized species such as *A.testuineus* are significantly higher compared to other fish such as *G. giuris*. On the other hand, Hossain and Halder (2015)[11] found that the median lethal concentration (LC50) of Talstar 2.5 EC on fry was 0.0014 ppm for 48 hrs which is very much similar to the findings of the present study (LC diazinon on *G. giuris*) although the exposure time was less for *Labeo rohita*. Thus during observation it was found that the effect of diazinon in the small fishes in their early stages are very vulnerable to the exposure of diazinon compared

to larger ones. Dutta and Meijer (2003)[12] reported that the age of fish, size, and duration of exposure affect the toxicity potential of diazinon in various fish species and the degree of its sensitivity varied even among the fish of the same genus and family. Rahman *et al.* (2002)[9] also reported that the LC50 value for 96 hrs of exposure to diazinon for the small fish *Barbodes gonionotus* to be 2.72 ppm which is much higher than the present findings. Phylogenetic difference between the species might be the reason for large variation in the LC50 value.

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Bioethanol production from renewable raw materials wheat straw and waste vegetables

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Abstract- Bioethanol significantly contributes to the reduction of crude oil consumption and environmental pollution and has been identified as the mostly used biofuel worldwide. It can be produced from various types of feed stocks such as sucrose, starch, lignocellulosic and algal biomass through fermentation process by microorganisms. *Saccharomyces cerevisiae* is the common microbes employed in ethanol production due to its high ethanol productivity, high ethanol tolerance and ability of fermenting wide range of sugars, compared to other types of microorganisms and yeasts. Yeasts can directly ferment simple sugars into ethanol while other type of feedstocks must be converted to fermentable sugars before it can be fermented to ethanol. The common processes involves in ethanol production are pretreatment, hydrolysis and fermentation. The efficiency and productivity of ethanol can be enhanced by immobilizing the yeast cells. This review highlights the different types of yeast strains, fermentation process, factors affecting bioethanol production and immobilization of yeasts for better bioethanol production.

Keywords:- ***Saccharomyces cerevisiae,*** **bioethanol, waste vegetables, wheat straw,**

INTRODUCTION

In today's modern society global economic growth contributes to a rapid increase in the consumption of traditional energy sources. According to numerous energy consumption analyses, the progressing depletion of fossil fuels calls for new initiatives on the market of renewable energy. Biomass today is regarded as one of alternative energy sources.[1,2] At present, only about 5 billion out of 150 billion tons of biomass harvested each year are processed into food. Biomass is not used for energy generation to the extent permitted by the existing technology.[3,4,5] Renewable energy sources today are becoming increasingly important in the energy balance of the country, and they are a characteristic feature of innovative and forward-looking economies (Kogut *et al.* 2012)[6]. There are various steps that can be enhanced for energy generation from biomass by combustion, gasication, ethanol and methanol fermentation or by using oilseed crops as a source of fuel. According to Nguyen *et al.* (2013)[7], energy generated from straw by gasification seems to be more environmentally-friendly than that produced by straw combustion. The combustion of fossil fuels produces harmful emissions to ambient air, mainly CO_2, which contribute to the greenhouse effect leading to global warming[8]. Thus the use of straw acts as an alternative source of energy that could reduce global

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warming and the depletion of fossil fuels[9,10]. The energy value of two tons of wood or straw is equivalent that that of one ton of high-quality hard bituminous coal.

One of the methods of generating energy from biomass is alcoholic fermentation. Simple sugars are converted into ethanol by yeasts[11-12] and these steps have been taken since generations. Ethanol is dehydrated and used to enhance or substitute Petroleum[13-14]. A variety of feedstocks from the first, second and third generation has been used in bioethanol production. Second-generation bioethanol comes from lignocellulosic biomass such as wood, straw and grasses. The first-generation bioethanol involves feedstocks rich in sucrose (sugar cane, sugar beet, sweet sorghum and fruits) and starch (corn, wheat, rice, potato, cassava, sweet potato and barley). The use of bioethanol can reduce our dependence on fossil fuels, while at the same time decreasing net emissions of carbon dioxide, the main greenhouse gas[1,2]. However, large-scale production of bioethanol is being increasingly criticized for its use of food sources as raw material.

Microorganisms such as yeasts play an essential role in bioethanol production by fermenting a wide range of sugars to ethanol. They are used in industrial plants due to valuable properties in ethanol yield (> 90.0% theoretical yield), ethanol tolerance (> 40.0 g/L), ethanol productivity (> 1.0 g/L/h), growth in simple, inexpensive media and undiluted fermentation broth with resistance to inhibitors and retard contaminants from growth condition. To obtain efficient ethanol fermentation with *S. cerevisiae*, numerous nutrients, including trace metals and vitamins, are required during the process[15-16]. Chemicals contribute significantly to the cost of large-scale production although it was not in the scope of this study to investigate this, their use should thus be minimized. Wheat hydrolysate, which is relatively cheap compared with chemicals, has been proven to be a potential supplement for lignocellulosic hydrolysate, because it is not only a sugar-containing material, but is also a complex nutrient source [17-18]. The production cost of ethanol is not only dependent on the yield but also on the concentration of ethanol in the fermentation broth, because of the high energy demand in the distillation step to obtain efficient ethanol from fermentation with *S.cerevisiae* numerous nutrients such as metals and vitamins are required during the process. Chemicals really lead to significant increase in the large scale production of ethanol. To obtain efficient ethanol fermentation with *S. cerevisiae*, numerous nutrients, including trace metals and vitamins, are required during the process. Chemicals contribute significantly to the cost of large scale production[20]; although it was not in the scope of this study to investigate this, their use should thus be minimize

MATERIALS & METHODS

Ethanol (also called ethyl alcohol, grain alcohol, drinking alcohol, or simply alcohol) is a chemical compound, a simple alcohol with the chemical formula C_2H_6O. Its formula can be also written as CH3"CH_2"OH or C_2H_5OH (an ethyl group linked to a hydroxyl group), and is often abbreviated as EtOH. Ethanol is naturally produced by the fermentation of sugars by yeasts or via petro chemical processes, and is commonly consumed as a popular recreational drug. The present project was conducted to evaluate sacchrification and fermentation (SSF), SPWS and presacchrified wheat meal of several cellulose material .

Raw Material used

Corn grain: Corn grain such as wheat straw and waste vegetables was obtained from the local markets and main sabzi mandi of Madhepura city and then stored at room temperature before pretreatment.

Saccharification and Enzymatic hydrolysis

The wheat straw was impregnated with a weak sulfuric acid solution and then squeezed using a hydraulic press. The pressed material was pretreated in a steam pretreatment unit and the whole slurry was then mixed with the PWM. The two materials were mixed in different proportions to investigate the effects on the ethanol yield and the ethanol concentration α-Amylase α-Amylase amylolytic were the major enzymes that were used for starch liquefaction and saccharification, respectively. The wheat straw was immersed in an aqueous solution of 0.2% H_2SO_4 at a liquid: dry straw weight ratio of 20. It was stored in sealed buckets for 1 h, and was then squeezed in a manual 3 L press to an average dry matter content of 43%. Steam pretreatment was performed in a unit described previously comprising a 10 L pressurized vessel, with a flash cyclone in which the pretreated material was released and collected. Previously optimized conditions for wheat straw were used; that is, the temperature was

maintained at 190°C for 10 min using saturated steam. Each batch that was fed into the reactor was 600 g wet weight. The steam-pretreated wheat straw (SPWS) was then subjected to SSF. The fermentor was loaded with the SPWS and the nutrients, which were sterilized separately at 121°C for 20 minutes, but the PWM was not sterilized because the starch and the enzymes already added would have been damaged at these conditions. SSF was performed at 36.5°C, and the pH was maintained at 5 ± 0.2 by addition of 10% NaOH solution. The experiments were run for 72 h in the case of pure PWM, and for 96 h in the case of SPWS or mixtures of the substrates. Saccharification of the raw material was conducted according to Guo *et al.* (2008)[20] with some modifications.

RESULT & DISCUSSION

The ethanol yields were calculated as a percentage of the maximal theoretical yield for glucose (0.51 g/g) that could have been produced if all the glucose present in the slurry and the PWM, including both monomers and oligomers in the liquid and glucan fibres in the WIS, had been converted to ethanol. The highest yield was obtained when equal amounts of PWM and SPWS were used. Thus, a mixed substrate is favorable in terms of final ethanol yield, probably due to the stress on *S. cerevisiae* caused by weak acids present in SPWS. At the same time, it is also easier to reach a high ethanol concentration using such as mixture than when using wheat straw only as a raw material.

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Studies on behavior of common baron *Euthalia aconthea*

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Abstract- : The objective of the present study was to identify the preference habitat (ecology), as well as host-plant interaction and different behavioral (ethology) patterns of *Euthalia aconthea*. The study was conducted in the campus of PG Department of Zoology Ranchi University, Ranchi. Since the species selected are rarely found therefore it was reason for selection of the species for research work.

Keywords:- Arthropoda, Lepidoptera, *Euthalia aconthea*, ecology

INTRODUCTION

Insects are the unique arthropod bearing head, thorax and abdomen, a single pair of antennae, three pairs of walking legs, one or two pairs of wings, and a long abdomen bearing distally the genital appendages.[1] Insects are the species that constitutes more than half of earth diversity[2]. And among the species butterflies are regarded as one of the best taxonomically studied group of insect[3]. India is described as a "Butterfly paradise" by Vankatramani[4]. Butterflies belong to the order Lepidoptera. Lepidoptera are the most familiar and easily recognizable of all insects and it is in this order that coloration has reached the highest degree of specialization[5]. They been studied systematically since early 18th century, so far 19,238 species documented worldwide of which, 1,504 species occur in India with 100 (15%) endemic and 26 (1.08%) globally threatened species in peninsular India, 334 butterflies species were reported from the Western Ghats and 150 species from the Eastern Ghats region[6]. Butterflies are scaled wing insects belonging to the order Lepidoptera of class Insecta[7]. Butterflies are regarded as one of the best indicator of the habitat quality. Worldwide there are more than 28,000 species of butterflies, with about 80 percent found in tropical regions. If we are to have any chance of effectively conserving insect populations globally, we must understand their ecologies[8]. Butterflies enable sustenance of ecosystem services through their role in pollination and serving as important food chain components. Being potential pollinating agents of their nectar plants as well as indicators of the health and quality of their host plants and the ecosystem as a whole, exploration of butterfly fauna thus becomes important in identifying and preserving potential habitats under threat. There has also been an alarming rise in industrial and automobile pollution in Indian metropolitan

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cities. *Euthalia aconthea* is recorded as minor pest infesting mango plantations. It is regarded as one of the most destructive species to mango plantations causing severe damage of the leaf. Regarding this the detailed study of the insect was done.

MATERIALS & METHODS

Study site

The study site was selected in the campus of University Department of Zoology Ranchi University located Jharkhand. The campus building is surrounded by garden with a large public maidan or field, and a nursery on the western region and constructed buildings on all other sides. The sprawling campus had good vegetation with different herbs, shrubs and trees. The western side of the campus building had wild plants growing along with cultivated plant species. The present study was conducted in a garden created on the south side of the campus to compensate for the habitat loss in the area due to construction and building activity.

Study period

The study period was divided into two phases. In the first phase, survey of butterflies habitat and behavioral activities within an unmanaged garden in the campus was done and in the second phase, setting up a garden in another site of same campus with plantation of host plants preferred by butterfly and caterpillar species (*Euthalia aconthea*) was done and observed for visits, host-plant preference and ethological activities. The study was conducted during the month of October 2018-February 2019.

Field survey for eco-ethology and behavioral study

A study on *Euthalia aconthea* one of the butterfly species was done in the college campus. A checklist containing butterflies in relation to host-plant preferences was given special preference. Different behavioral activities of butterflies were studied using focal-animal sampling method[9-10]. Time duration was also calculated. Behavioral allocation in man hours of different activities in different species of butterflies was done over a period of 5 months (Two hours in the morning and Two hours in the evening with intermittent gaps, October -December). Observation was done specially keeping attention the behavioral pattern, host plant preference etc. Each individual of *Euthalia aconthea* was observed for 5 minutes. During this sample period, all the different activities by the individual were recorded. A time-budget of behaviors like foraging, nectaring, flower-visiting (flight), defending attitude, resting, courtship flight, was also recorded.

RESULT & DISCUSSION

During observation it was found that most of the *Euthalia aconthea* species habituated in the mango tree and completed its half of the life cycle such as from larval



Fig.1- *Euthalia aconthea*

to pupal stages. Like other members of the order Lepidoptera *Euthalia aconthea* goes through four different stages to complete its life cycle.egg, larva, pupa and adult.

Mating behavior

Mating in adult stage of *Euthalia aconthea* was observed during the flight hours and usually lasted for 70-80 minutes. The eggs were laid under the leaves surface of the mango tree in single clutch. The eggs were hemispherical in shape with base diameter of about 1.5-1.8 mm. The surface of the eggs was covered with hair like structures. The fresh eggs were green in colour but later on it turned into darker.

Incubation Period

The incubation period lasted for about 4-5 days. From the incubated eggs young larva (*Mangifera indica*) hatched out. The young larva usually fed on fresh mango leaves.

Larval Stages

Five different larval periods was observed during the investigation period, first instar, second instar, third instar, fourth instar and fifth instar. The larval stage of *Euthalia aconthea* had hair like structures that surrounded the entire body and was generally used as defensive organ.

Pupal stage

Fully fed and matured larva feeding and started to spin silk thread around itself to make silk mound. The prepupal larva hangs itself from the anchor pont in a head down posture. The pupal period lasted for 7-8 days prior to adult stage.

Adult stage

After 7- 8 days of pupal stage the adult larva emerged out from the pupa. The adult larva was brownish in colour with brown antennae head, thorax, abdomen etc. the proboscis was lime green in colour. The wings were pale brown in colour with black spots with expansion of about 68-79 mm.

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Studies on callus induction of *Murraya koenigii* Spreng in auxin and cytokinin

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Abstract : Different concentration of phyto-hormones affected callus formation of *Murraya koenigii* Spreng was cultured in MS Medium supplemented with different concentration of 2, 4-D, IAA, IBA, NAA, KN and coconut water. Cultures were kept on 25 ± 2^{0}C temperature and 16 hour photoperiod while callus was observed on different counteraction of auxin or cytokinin alone or in combination. Most suitable medium for callus formation from shoot tip, leaf internode was that 2,4-D (2.0mg/l), IAA (2.0mg/l), IBA (2.0mg/l), NAA (2.5 mg/l), 2,4-D (2.5mg/l) + IBA (1.5mg/l) + 20% CW, 2, 4-D (1.5mg/l) + KN (0.5mg/l).

Key words: *Murraya koenigii*, auxin, cytokinin, culture

INTRODUCTION

Murraya koenigii Spreng is commonly known as "curry leaf" belongs to the family Rutaseae. It is found in outer Himalayas, from the Rasi eastwards, ascending to 5,000 feet in Assam, Chittagong upper and lower Burma. It is evergreen deciduous small shrub. It reaches a maximum height 2.5 meters high. The main stem is dark green to brownish, with numerous dots on it. The girth of the main stem in 16cm. leaves estipulate, bipinnately compound 30cm long. Each bearing 24 leaflets, having veticulate venation, lanceolate, 4.9 cm long, 1.8 cm broad, having 0.5 cm long petiole. Plant tisane culture technique provides a system for rapid production of large number of genetically uniform disease resistant plantlets during the past few decades. Soon it was discovered that this new area of plant biology had practical value for commercial, medicinal and agricultural plant propagators as well as in conservation of valuable plans[1,2].

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Culture Media: MS Medium with varying concentration of phytohormones was employed for callus initiation.

Culture Condition: The cultures were maintained under controlled environment at 25 ± 2^{0}C with 16 hours photoperiod and 8 hours dark. All experiments were conducted under sterile conditions.

MATERIALAND METHODS

In the present analysis leaf and leaflets were used for experiments. The segments of leaves were inoculated in MS medium supplemented with different phyto hormones and combinations.

The combination and concentrations of hormones were used Table – I.

RESULT AND DISCUSSION

In *Murraya koenigii* Spreng, induction of callus was achieved by supplementing low concentration of auxin. In MS + 2, 4-D (2.0mg/l) 80%, MS + IAA (2.0mg/l) 65% callus, MS+IBA (2.0mg/l) 75% callus, MS + NAA (2.5mg/l) 55% callus, MS+2, 4-D (2.5mg/l) + IBA (1.5mg/l) 20% CW 91% callus, MS + 2, 4-D (1.5mg/l) + KN (0.5mg/l) 81% callus was observed.

After 4-5 weeks in when callus was transferred. Some works reported on high frequency shoot regeneration from *in vitro* raised seeding of *Murraya koenigii* Spreng. The cotyledonary node explants were cultured on MS medium alone or supplemented with IBA, KN and 20% coconut water.

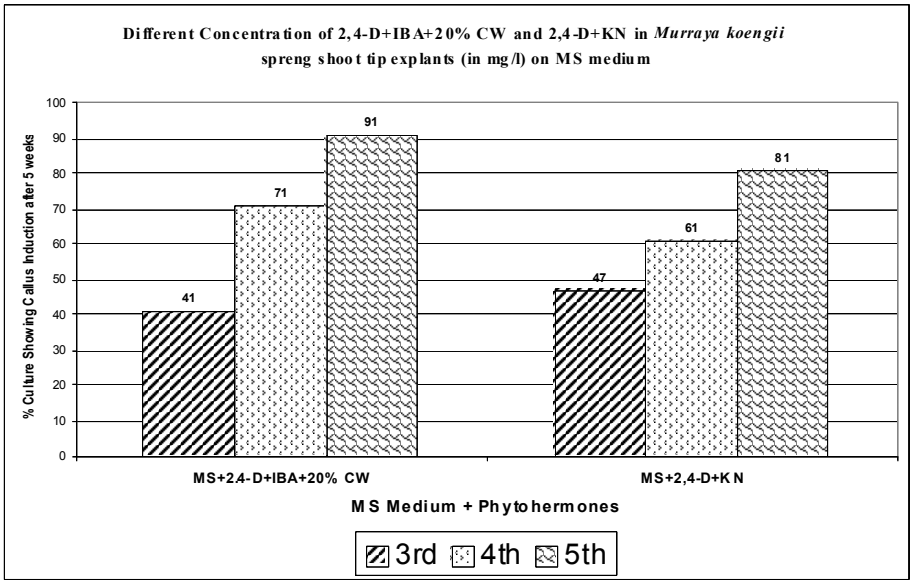
In other plant of family Rutaceae tissue culture studies and regeneration of shoot tip and leaf internode explants via organogenic callus culture were achieved.

The callus formation when basal median was supplemented with 2, 4-D ranging from 0.5mg/l to 5.0mg/l for sugarcane.[3] He also stated that callus formed at lower concentration of 2, 4-D was whitish and loose while higher concentration was brownish.

The multiple shoot bud originated directly from epidermal and sub-epidermal layers of apical bud explants similar results have been reported in *Capsicum annuum* L. and *Acalpha wilkesiana* 'Dwarf'.[4,5] There are several reports of regeneration of *Albizzia lebbeck*.[6] It has been reported regeneration of complete plants by inducing adventitious shoots in the cultures of immature embryos.[7] Several attempts have been made to utilize the *in–vitro* micropropagation method for rapid clonal multiplication of many tropical subtropical and temperate fruit.[8]

Table 1. Callus initiation (*Murraya koenigii* Spreng.) in different phytohormones incorporated in MS median.

Phytohormones concentration (in mg/l)	Name of explants	% initiation of the callus of *Murraya koenigii* Spreng.		
		3rd week	4th week	5th week
MS + 2,4-D (2.0mg/l)	Shoot tip, leaf internode	20%	45%	30%
MS + IAA (2.0mg/l)	Shoot tip, leaf internode	35%	45%	65%
MS + IBA (2.0mg/l)	Shoot tip, leaf internode	35%	46%	75%
MS + NAA (2.5mg/l)	Shoot tip, leaf internode	25%	35%	53%
MS + 2,4-D (2.5mg/l) + IBA (1.5mg/l) + 20% CW	Shoot tip, leaf internode	41%	71%	91%
MS + 2,4-D (1.5mg/l) + KN (0.5mg/l)	Shoot tip, leaf internode	47%	61%	81%



Regeneration of adventitious shoot buds only noted in case of if the leaf explants are collected from the vegetative phase i.e. prior to flowering leaf explants of *Echeveria* species that are collected from young leaves produced only root.

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Ethnobotany and the Sacred Divine Plant: Tulsi

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Abstract : Ethno botany is the branch of Botany that studies the relationship of the plants with the people and the relevance of the plant in cultural practice, religious values and medicinal values for curing disease. *Ocimum Sanctum* is most sacred plant in India, commonly found in most of the Hindu household due to its religious profile. No plant in any part of the world commands such respect. Commonly known Tulsi, *Ocimum sanctum* is a wonder herb due to its tremendous medicinal values. The chemical composition of Tulsi is highly complex and contains many nutrients and biological compound, makes it great medicinal herb used in India since time immemorial for curing and preventing disease.

Key words: *Ocimum Sanctum*, medicinal values, Sacred

INTRODUCTION

Ethno botany: Ethno botany is the scientific study of the relationship between people and plants. It is a branch of Botany concerned with the use of plants in folklore, religion, culture etc. The word ethno is derived from ethnic that refers to people, culture, beliefs, aesthetic, knowledge and practice. Botany is the study of plant kingdom, be it a tiniest fern or blade of grass or the tallest and greatest tree. Ethno botanical knowledge encompasses species and is rooted in observation, relationship, needs and traditional ways of knowing. Ethno botany is an integrative multidisciplinary field of learning. Therefore the tools of ethno botanical investigations are many, botany, mycology, taxonomy, anthropology, ethnography, archaeology, comparative folklore, religious studies, chemistry, medicine, pharmacology and more.

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The term 'ethno botany' was first applied by Harshberger in 1895 to the study of plants used by primitive and aboriginal people. Kirtikar and Basu (1935) stated, "The ancient Hindus should be given the credit for cultivating what is now called ethno botany". Jones (1941) defined it as the study of the inter-relations of primitive man and plants. Schulte (1962) interpreted ethno botany as "usually the study of relationship which exists between people of a primitive society and their plant environment". Jain (2001) laid down that "ethno botany deals with the study of total natural and traditional inter relationships between man and plants and his domesticated animals". Ethno botany must have the first knowledge, which the early man had acquired by sheer necessity, intuition, observation and experimentation (Bedding, 1925; Apparanathan and Chellandurai, 1986; Battacharyya, 1988 and Jain, 1997). Since Harseberger the definition of ethno botany has changed & evolved along with the formation & evolution of the field (Jain, 1996).



Ocimum sanctum (Tulsi) :

Kingdom	: Plantae
Class	: Angiosperms
Order	: Lamiales
Family	: Lamiaceae
Genus	: *Ocimum*
Species	: *O. sanctum*

Ocimum sanctum (Holy basil) is an erect, many-branched sub-shrub, 30–60 cm (12–24 in) tall with hairy stems. Leaves are simple opposite green or purple, petioled, with an ovate, up to 5 cm (2.0 in)-long blade which usually has a slightly toothed margin; they are strongly scented. The purplish flowers are placed in close whorls on elongate racemes. Tulsi is native throughout the world tropics and widespread as a cultivated plant and an escaped weed. It is cultivated for religious and medicinal purposes and for its essential oil. Tulsi is an important symbol in many Hindu religious traditions, which link the plant with Goddess figure. The name 'Tulsi in Sanskrit means 'the incomparable one'. The presence of a Tulsi plant symbolizes the religious bend of a Hindu family.

Local names- Hindi: *Kala Tulsi*, English: *Holy Basil,* Malayalam: *Trittavu*, Marathi: *Tulshi*, Telugu: *Tulsi*.

Sacred Aspects and Religious Profile: "The Tulsi is most sacred plant in India. No Plant in the world commands such... universal respect, adoration and worship from the people as does Tulsi. It is the plant par excellence." [Indian Botanical Folklore] "A house with a Tulsi plant in front of it is a place of pilgrimage... The wind that carries the aroma of Tulsi spreads purity wherever it blows."

In India, many traditional Hindus grow Tulsi and have at least one living Tulsi plant. They use its leaves in routine worship; they feel protected by its sacred aura; and they use rosary beads for meditation made from its cut stems.

The origin of Tulsi is cited in Padma Purana. As per the verse in Padma Purana Lord Brahma has narrated the story of origin of Tulsi to sage Vasistha. As per the story there was a ruler of Mathura a demon named Kalnemi. Her daughter named Vrinda was very beautiful and got married with King Jalandhar, the demon of water. She was completely devoted to her husband and therefore created a shield around him that made him invincible. Jalandhar claimed himself the Lord of the oceans and when the ocean chummed in the search of immortal nectar numerous valuable things were obtained. He claimed all the things being supposed himself the Lord of ocean. This created the conflicts between him and god. But he was invincible as long as Vrinda stood beside him. To overcome his power Lord Vishnu charm Vrinda in form of Jalandhar and was successful to take her out of the grasp of the demon, which led to the defeat of Jalandhar. When Vrinda came to know about the trick, grieved by her husband death, she cursed Lord Vishnu to become a black stone called shaligram and also cursed him to lose her wife Lakshmi. Lord Vishnu blessed her to become Tulsi plant for her devotional nature and also promised her to marry her after year on the day of Krishna ekadashi.

Srimad-Bhagavatam explains the special position of the Tulsi plant: "Although there are numerous flowering plants full of transcendental fragrance in the spiritual realm, they are aware that Tulsi is given special preference by the Lord, who garlands Himself with Tulsi leaves." Satyabhama once weighed Lord Krishna against all her legendary wealth. The scales did not balance until a last single tulsi leaf was placed along with the wealth on the scale by Rukmini with devotion. Thus, tulsi played the vital role of demonstrating that even a small object offered with devotion is of greater value than all the wealth in the world. Worship of Lord Vishnu and his one of the most famous incarnation Shri Krishna is incomplete without offering Tulsi leaves. Devotees of Lord Krishna chant on beads made from the wood of the Tulsi plant and wear Tulsi beads around their necks. There is a Sanskrit: "Tulanaa naasti athaiva tulsi" that which is incomparable in its qualities is the tulsi. It is the only pooja samagri which can be washed and reused. A traditional prayer tells that the creator-god Brahma resides in its branches, all Hindu pilgrimage centers reside in its roots, the Ganges flows through its roots, all deities in its stem and its leaves and

the Hindu scriptures - the Vedas in the upper part of its branches. It is called as "the central sectarian symbol of Hinduism" and Vaishavas consider it as "the manifestation of god in the vegetable kingdom". Garlands made of 10000 tulsi leaves, water mixed with tulsi, food items sprinkled with Tulsi are offered in veneration to Vishnu or Krishna.

A ceremony known as Tulsi Vivah is performed by the Hindus between Prabodhini Ekadashi (eleventh lunar day of the waxing moon of Kartika) to Kartik Poornima (full moon in Kartika), usually on the eleventh or the twelfth lunar day. It is the ceremonial wedding of the Tulsi plant to Vishnu, in the form of his image, Shaligram or a Krishna or Rama image. Both the bride and the groom are ritually worshipped and then married as per traditional Hindu wedding rituals. It marks the end of the four-month *Chaturmas* period, which corresponds to the monsoon and is considered inauspicious for weddings and other rituals, so the day inaugurates the annual marriage season in India.

Phytochemistry of Tulsi: The leaves of Tulsi contain volatile oil comprising about methyl eugenol, eugenol, carvacrol and sesquiterpine hydrocarbon caryophyllene. The oils of *Ocimum sanctum* carries five fatty acid named – stearic, palmitic, oleic, linoleic and linolenic acids. It also contains a number of sesquiterpenes, monoterpenes, viz. bornyl acetate elemene, neral and pinenes, camphene, campsterol, cholesterol, stigmasterol and sitoterol. It is a good source of beta carotene, calcium and vitamin C. It also contains tannins, camphor, flavanoid, triterpene, urolic acid, zinc, maganese, and sodium. Essential oils of Tulsi are antibacterial, antifungal and antiviral. The root contains beta-sitosetorol and three triterpenes A, B and C.

Ayurveda and Tulsi: - Ayurveda is supposed to be a science of life and the world's oldest medical system. It has a holistic approach to health and disease that focuses on preserving and promoting good health and preventing disease through healthy lifestyle practices. These practices include consumption of fresh, minimally processed foods, the use of chemical formulas that eradicate ageing and disease, sophisticated detoxification practices and regular consumption of adaptogenic herbs that enhance the body's capacity to maintain balance in the midst of a variety of stressors. Ayurveda's use of medicinal and culinary herbs draws upon India's incredible biodiversity with a variety that is unsurpassed by any medical system; yet, of all the herbs used, none has a status comparable to tulsi or holy basil (*Ocimum sanctum*). Very commonly found in India, Tulsi is recommended for hundreds of serious disorders and is actually highly recommended as daily prophylactic to prevent disease. Tulsi is pungent and bitter in taste (*rasa*), pungent in post digestive effect (*vipaka*) and has hot potency (*virya*). It alleviates *kapha* and *vata dosas*, but slightly aggravates the *pitta* dosa. It possesses light and dry attributes. On the contrary the seeds are oily (*snigdha*) and slimy (*picchila*) in attributes and have a cold potency. Tulsi is a stimulant, aromatic herb and effectively reduces the fever. Ayurvedic wisdom, suggests that tulsi is a tonic for the body, mind and spirit that offers solutions to many modern day health problems. Tulsi is credited with giving luster to the complexion, sweetness to the voice and fostering beauty, intelligence, stamina and a calm emotional disposition.

Tulsi: Medicinal uses: - The seeds, leaves and roots of the Tulsi have great medicinal value. It is used both internally and externally. Some of the medicinal uses of Tulsi are as mentioned below:

Relives Fever:- Highly potent anti bacterial, antibiotic, germicidal and fungicidal properties are present. The leaves of Tulsi is very helpful in alleviating or curing almost all types of fevers. Fever is mainly caused due to infections from protozoa (malaria), bacteria (typhoid), viruses (flu), and even allergic substances and fungus. The tremendous disinfectant, germicidal, and fungicidal properties of tulsi destroy all those pathogens discussed above and heal the resulting fever. It is an old practice in India to have a decoction of tulsi leaves and flowers in case anyone is suffering from a fever.

Dental Care: Holy basil destroys the bacteria that are responsible for dental cavities, plaque, tartar, and bad breath, while also protecting the teeth. It also has astringent properties which make the gums hold the teeth tighter, thereby keeping them from falling. However, tulsi also has certain compounds like mercury, which has rich germicidal properties that can be harmful to the teeth if kept in direct contact for too long. Therefore, the leaves when chewed mitigate the infections of gums.

Kidney stone: Being a great detoxifier, an effective pain-killer, and a mild diuretic, Tulsi is also very effective in easy dissolution of kidney stones, and expelling the same painlessly through the urinary tract. It also helps clean out

the kidneys through increased frequency of urination. Acetic acid and certain components in Tulsi essential oil facilitate dissolution of the stones. Finally, it has pain-killer effects and helps bear the pain from kidney stones as they pass. To cure the kidney stone it is suggested to take a mixture of juice of Tulsi leaves and honey, everyday for approx six months regularly.

Skin Care: The presence of essential oils in Tulsi, which are highly antibiotic, disinfectant, antibacterial, and antifungal in nature a smooth paste of tulsi leaves on skin rashes, acne, pimples, wounds, etc., is immensely useful for curing the same. External application on the skin also removes extra oil from the surface of the skin. Camphene in it gives a soothing and a cooling effect. The message with the leaves juice improves the circulation be beneath the skin augments the sensation in the skin.

Eye Care: Regular consumption of leaves of Tulsi can protect eyes from the damage done by free radicals, such as cataracts, macular degeneration, glaucoma, vision defects, and ophthalmia, due to the high antioxidant content of its essential oils, vitamin A, and vitamin C.

Insect bites and stings: The prophylactic properties of Tulsi leaves are very supportive for curing bites and stings of insects. For this purpose, the affected areas must be fed with a teaspoonful of juice taken from fresh leaves of Tulsi, regularly after few hours. A paste of the fresh roots of Tulsi may instead be applied on the affected areas.

Helps in treatment of Cough, Cold and Asthma:- Tulsi has specified actions on the respiratory system-*pranvaha srtosa*.It effectively liquefies the phlegm due to its hot and sharp attributes. It gives excellent result in cough due to *Kapha,* allergic bronchitis, asthma and eosinophilia. Combined with honey, the juice works well to control the hiccup. In tubercular cough, tulsi is also beneficial. A tea prepared with the leaves of tulsi is a common domestic remedy for cold, cough, mild indigestion, diminished appetite and malaise.

Beats Stresses and Pains: Leaves of Tulsi do possess adaptogen (stress relieving) properties along with pain-killing properties. Hence, leaves or juice of Tulsi help(s) in soothing the nerves, lowering the blood pressure and in regulating blood circulation, in beating the damaging effects of free radicals, and thus, in reducing inflammation/ pains and stresses in the body. For beating stresses of daily life, it is suggested that one should consume 10-12 leaves of Tulsi every day.

Prevents Lungs disorder:- The compounds like vitamin C, camphene, eugenol, and cineole present in the essential oils of tulsi not only provide relief from the infections in the lungs but also help clear the lung congestion. Furthermore, they are found to be effective in healing the damages caused to the lungs due to smoking etc.

Prevents Heart disease:- It is a good blood purifier and is helpful in heart disease. Vitamin C and other antioxidants such as eugenol protect the heart from the harmful effects of free radicals. Additionally, eugenol is very beneficial in reducing cholesterol levels in the blood.

Chemo preventive properties: - Tulsi may inhibit the growth of various cancers including oral cancer due to the presence of phytochemicals like eugenol.

MATERIALS AND METHODS

Several research papers have been consulated.

CONCLUSION

Ocimum Sanctum is one of the holiest herbs found in India and posses high religious and immense medicinal value. Due to the presence complex chemicals and bioactive compound Tulsi (*Ocimum sanctum*) may be used in various diseases in raw form and valuable medicinal drugs may be prepared from the all parts of it.

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